

Multi-Grid TBox Network

Routing, Virtual Circuits and Adaptive COSTK

Detailed architecture, interfaces, implementation phases and verification specification

Revision 7 | Engineering design | 6 September 2026

Purpose

A conventional engineering specification for building a routed, multi-grid TBox network with optional explicit MPLS paths, stable-IP mobility and access-edge TCP adaptation. Decisions, mechanisms, implementation work and release evidence are separated throughout.

Status

Proposed design baseline for review and staged implementation. This revision is not a claim that GVC, mobility, kernel integration or the acceptance tests have been implemented or passed. Vendor and hardware gates remain explicit.

Requirements basis: Integration_Request_Final.docx and 20260831_1. v1_v2
Grid_TBox_EasyMesh_Implementation_Plan.docx.

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1. Purpose, requirements and design status

1.1 Purpose and intended use

This document specifies a network built from multiple TBox grids, relay TBoxes, constrained mesh access routers and Internet gateways. It describes how to separate radio management from routing, add connection-oriented paths without replacing the IP underlay, preserve mobile IPv4 sessions across access changes, and apply COSTK only where an access-edge TCP proxy is useful. The intended readers are the platform, radio, routing, COSTK, controller and test engineers who will implement and qualify these functions.

The document is an engineering baseline rather than a presentation. Its chapters define responsibilities, data models, packet handling, state transitions, failure behavior and implementation gates. Section 12 gives the ordered work packages; section 13 defines the corresponding evidence. An engineer should be able to identify what to build next, what must already work, and which observation permits progression. A test specification is not a test result.

Revision 6 replaces the previous slide-style, landscape design. It integrates corrections into the relevant chapters instead of preserving contradictory older statements and adding another appendix. In particular, it replaces the earlier segment-stack example, unspecified service-label context, automatic MPLS/Babel fallback and implicit fast-protection claims with an explicit, capability-gated implementation baseline. The archived Revision 5 is historical material, not an alternative implementation specification.

Revision 7 adds explicit scalability treatment for EasyMesh managing the **5 × 5, 25-TBox grid itself**. This is not merely a limit on household mesh extenders. Sections 2.6–2.10 preserve the physical grid while defining qualified controller capacity, optional bounded radio-management cells, their routed boundaries and the consequences when the vendor cannot provide that separation. The phase plan and E01–E10 tests make this a release gate rather than an assumed benefit of Babel.

1.2 Governing inputs and interpretation

The governing input documents are **Integration_Request_Final.docx** and **20260831_1.v1_v2 Grid_TBox_EasyMesh_Implementation_Plan.docx**, supplied from the user's Downloads directory. The former identifies eleven integration questions and constraints; the latter describes the grid topology, v1 EasyMesh arrangement, v2 routing direction, addressing and vendor questions. The supplied grid drawing provides topology context: repeated 5 × 5 grids, relay exits and a long-distance point-to-point backbone.

The inputs are requirements and proposals, not proof of hardware behavior. Their desired 5–7 Gbit/s node capacity, approximately 10 Gbit/s long-distance backbone, multi-direction operation, self-healing and low degradation must be converted into measured acceptance criteria on actual hardware. No document statement establishes radio coexistence, vendor API availability, hardware MPLS offload or seamless roaming by itself.

Subsequent discussion adds native MPLS carriage of IP/TCP, stable-IP mobility, TCP control-packet treatment, pre-association, retained COSTK owners and a staged

implementation plan. These are design refinements. They are not attributed to the original requirements where those documents did not specify them.

In this document, **required** means necessary for the selected design profile; **proposed** means software or configuration that still needs implementation; **qualified** means an identified hardware/software combination passed the named tests; and **deferred** means excluded from the initial release. Configuration examples, limits and addresses are design examples, not instructions to modify a live network without an approved deployment manifest.

1.3 Scope and initial release decisions

The initial routed release uses standard Babel for ordinary within-grid reachability, explicit controller-installed hop-by-hop MPLS where a circuit is requested, and standard IPv4 service tunnels where a mobile or proxy packet must reach a particular processing context. EasyMesh remains responsible for supported radio/link management. A regional routed backbone connects multiple grid domains to gateways. It must not depend on extending every subscriber bridge across the region.

Stable mobile addressing is initially IPv4. A mobile has a home gateway anchor and a mutable current access-edge binding. A COSTK flow additionally has an immutable proxy owner. These objects are independent: changing the access TBox does not transfer TCP socket state, change a live flow from COSTK to bypass, or move its NAT mapping to a new gateway.

The initial MPLS encoding uses one locally allocated transport label at each hop. A separately qualified service label may be considered later; it is not needed by the initial IP-in-IP service-context design. Segment routing, Binding SIDs, multi-tenant overlapping-address VRFs, packet replication/elimination and hardware fast-protection groups are separate extensions. They must not be prerequisites hidden in the baseline implementation.

The IP-in-IP service backend is a deliberate refinement of the earlier native-carriage discussion. Plain IPv4/TCP can be carried directly by MPLS. However, the selected mobile service must also distinguish access-origin, proxy-input, proxy-server-leg and proxy-client-return traffic. Dedicated trusted IP-in-IP endpoint pairs provide that context using standard facilities without adding UDP. This adds a 20-byte outer IPv4 header in the selected no-options profile; it is not a claim that MPLS requires IP-in-IP. Section 8 defines the contract and the alternatives that remain deferred.

No COSTK instance is installed on the memory-constrained leaf mesh router. No transit or strong-signal backhaul link runs an additional COSTK proxy. The leaf retains the vendor AP/EasyMesh functions and the existing small C telemetry reporter. A capable serving TBox makes the policy decision for new mobile flows, subject to the anchor's ownership admission barrier. Bypass retains the actual sender's congestion-control algorithm; CUBIC is a controlled experiment configuration, not something a bypass router can impose on an arbitrary Internet server.

1.4 Requirements traceability

The identifiers R1-R11 correspond to the eleven requests in Integration_Request_Final.docx. This table is the implementation handoff map. Detailed procedures and observables are in sections 12-13; an entry here is not evidence that the requirement has passed.

Requirement	Design disposition	Delivery and verification
R1: preserve EasyMesh responsibilities	Radio control remains with EasyMesh; Babel owns routed reachability; GVC owns explicit path installation. No competing radio controller is introduced.	Phases 0-1: record ownership, prove radio/L3 coexistence, recover link and whole-node failures.
R2: implement multi-link lifecycle	Introduce per-neighbor/per-link records with Discover → Validate → Measure → Stabilize → Advertise. Unknown metrics remain unknown.	Phases 0-2: field capability matrix; stale, reset, peer-add/remove and multi-link tests.
R3: flow-to-path service	GVC controller computes paths from measured topology; agents install local forwarding. A geographic address is not the path algorithm.	Phases 2-3: deterministic path tests, admission model, transaction replay and packet-path verification.
R4: multi-peer relay chain	Model every usable routed relay adjacency; remove single-peer assumptions in proposed collectors and path models, not by disabling security restrictions blindly.	Phases 1, 7-8: simultaneous peers, relay chain and bottleneck tests; one adaptive policy owner per access segment.
R5: whole-node failure	Use already usable alternative adjacencies; Babel recomputes IP paths and GVC repairs explicit state by its own mechanism.	Phases 1, 4, 7: power off node 2 and verify the actual alternative route, loss and interruption.
R6: MaxLinear metric feasibility	Per-MLO RSSI/SNR, retry/failure, queue and probe semantics require vendor evidence and measured validation.	Phase 0 inventory and Phase 1 collector qualification; missing fields may reduce capability but cannot be fabricated.
R7: unambiguous addressing	Reserve fixed access, link, loopback, service-endpoint and stable-mobile ranges in one IPAM authority. Reject overlaps before enrollment.	Phases 0-1 and 5: automated overlap/duplicate tests, lease and mobile-host-route tests.
R8: dual relay exits	Require physically/radio-diverse exits for a protected grid; record shared power, mast, channel and upstream risks.	Phases 4, 7-8: separate link/node/shared-risk failures; terrain exceptions explicitly reduce protection claims.
R9: enforce no feedback	COSTK exports lifecycle facts and policy observations without route-writing credentials. Metrics and route APIs reject COSTK-origin forwarding updates.	Phases 2 and 6: negative authorization tests and audit of routing changes during policy oscillation.

Requirement	Design disposition	Delivery and verification
R10: align deployment stages	Stage A is bench/emulation and Phases 0–6; Stage B is the single physical grid in Phase 7; Stage C is regional Phase 8.	Signed phase evidence and promotion decisions; later scale tests do not retroactively validate an earlier failed gate.
R11: routable interface coexistence	Vendor-supported per-link or equivalent isolated routed adjacencies must coexist with EasyMesh's actual backhaul interfaces.	Phase 0 blocking proof; if unavailable, retain qualified EasyMesh-only v1 rather than claim unproven Babel fallback.

Additional user requirement **ES1** concerns EasyMesh scaling across the physical 5×5 grid. Sections 2.6–2.10 and 6.4 define the selected mitigation and qualification envelope; section 12.10 assigns implementation work, and E01–E10 in section 13.5 define scale evidence. ES1 supplements rather than rennumbers the original eleven requests.

1.5 Evidence and implementation status

Repository inspection establishes that COSTK and its lightweight telemetry path exist. It does not establish that the proposed GVC controller, GVC agent, mobile binding service or first-SYN ownership admission service exists. These components are new work in this design. Proposed package names and API names describe contracts, not installed binaries.

The captured configuration at `build/edge-4.4.60/kernel.config` is evidence about that saved kernel build, not a remote read of every current TBox. It has MPLS core support but lacks routing and lightweight-tunnel configuration needed for the complete baseline. It also lacks the captured IPIP, NFQUEUE and TPROXY options discussed in sections 7 and 10. An exact target inventory and matching firmware build are necessary; merely copying a module from another OpenWrt release is not a supported solution.

The current relay EOF behavior in `src/proxy.cpp` also does not demonstrate the required TCP half-close semantics. That finding comes from code inspection, not a reproduced live failure. Phase 6 explicitly includes prequalification and repair before COSTK mobility is released. The routing and bypass-mobility phases can proceed independently of that repair.

This revision includes no newly executed radio, throughput or mobility measurements. Its validation covers document construction, traceability and logical consistency. Historical test reports remain separate evidence and must not be represented as proof of the new MPLS/mobility architecture.

2. System architecture and responsibility boundaries

2.1 Repeated grid and regional architecture

A grid is a routed administrative domain containing access-capable TBoxes and relay-capable TBoxes. The 5×5 layout is a useful deployment example, not an assumption that every drawn edge is usable or that every grid contains exactly 25 nodes. The topology database contains observed directional adjacencies and their shared-risk relationships. It does not infer a radio link merely because two nodes are adjacent on a map.

Each protected grid has two diverse relay exits. These connect to a routed regional backbone and one or more Internet gateways. A relay may serve multiple peers and multiple grids, provided its link capacity, forwarding resources and failure exposure are included in admission. Two exits on the same radio, channel, mast, power feed or downstream bottleneck do not constitute complete failure independence.

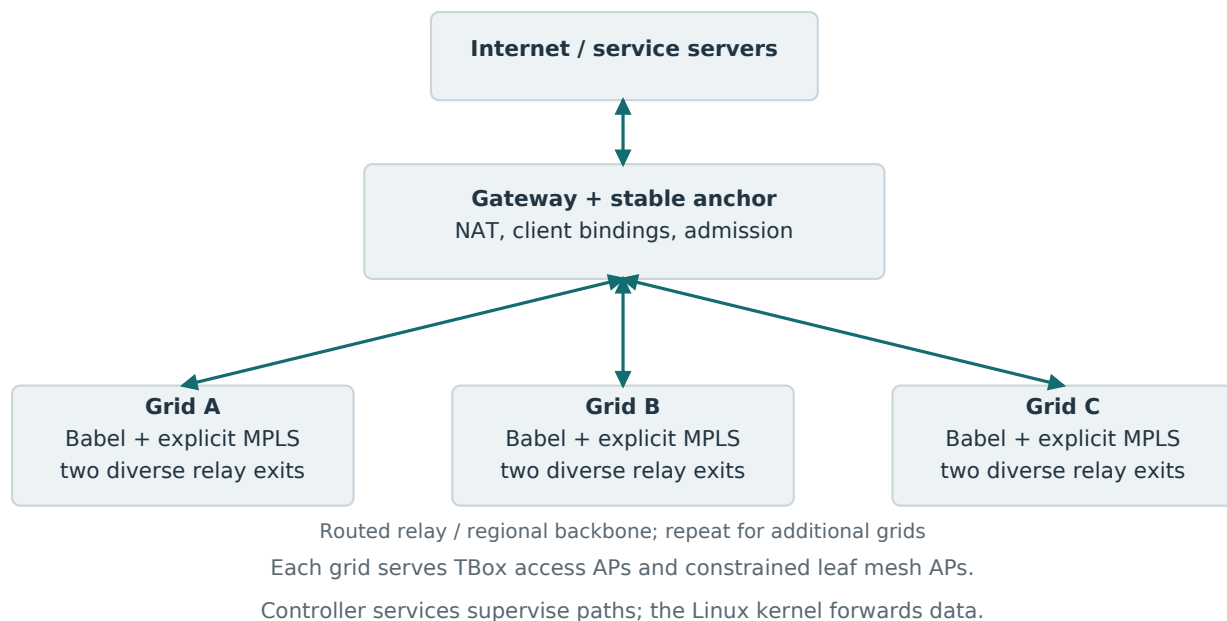


Figure 1. Repeated grid domains connect through relay exits to a gateway anchor. The drawing shows logical services, not a required single physical gateway.

Within a grid, Babel provides the ordinary IP underlay. For the initial regional design, grid boundaries exchange filtered aggregates and infrastructure reachability through eBGP or a separately operated routed backbone. The chosen deployment manifest identifies the actual regional implementation; it must not run mutually redistributing Babel and BGP defaults without filters and ownership. GVC can span a sequence of grid and regional domains only where each segment has a confirmed forwarding backend and an admitted path.

The regional controller sees border endpoints, measured domain reachability, capacity envelopes and shared risks. A grid controller maintains local link detail. A later hierarchical deployment may delegate subpath computation to a grid controller, but the initial control transaction still requires one authoritative circuit owner and explicit acknowledgments from

the affected domain agents. There is no unbounded national link-state flood and no claim that Babel itself supplies a complete traffic-engineering graph.

At first deployment, one active GVC/mobility controller can manage the laboratory domain. This is distinct from the EasyMesh radio-management controller assigned to each qualified cell and from a proposed regional management service. Production requires fenced, durable leadership and bounded controller-outage behavior for each relevant role. A hot standby without enforced single-writer fencing is not high availability. Retention of installed data-plane state during a GVC outage does not prove radio/link survival after EasyMesh-controller loss; that behavior is separately qualified. No edge elects an independent, conflicting mobile binding.

2.2 Component responsibilities

Component	Owns and produces	Must not do
EasyMesh / vendor radio manager	Secure onboarding, usable radio/backhaul links, supported channel and spectrum management, association events and supported VLAN/MLO operations.	Delegate regulatory decisions to an unqualified routing script or let two control domains fight over one radio.
Babel and regional routing	Ordinary prefix reachability, routing protocol state, selected IP next hops and permitted route exports.	Assume an installed MPLS next hop changes when the IP FIB changes.
Multi-link collector / metric adapter	Versioned link observations, validity and capability records; damped, supported metric inputs.	Convert missing retry data into zero loss or feed COSTK throughput scores back into route cost.
GVC controller / PCE	Path requests, constraints, capacity ledger, circuit generations and authenticated desired state.	Forward packets in userspace or treat the controller's graph as proof that an unformed RF link exists.
GVC agent	Local validation, prepare/readback/commit/drain/reconcile, labels and authorized forwarding backend operations.	Choose global policy independently, invent kernel capabilities or install unscoped customer forwarding from arbitrary callers.
Mobile anchor / mobility control	Stable lease authority, current-edge binding, authenticated handover transaction, anchor NAT and flow admission authority.	Change a live flow's immutable proxy owner on an attachment change.
Access agent and COSTK	Authorized attachment facts; local access policy for new flows; proxy sockets, lifecycle and resource reporting.	Act as a general backhaul proxy, write Babel metrics or migrate TCP state by changing a label.
Constrained leaf mesh router	Existing AP/EasyMesh service and small C telemetry reports to its serving TBox.	Run COSTK, GVC, a heavyweight new mobility agent or an experimental kernel dataplane.

GVC means Grid Virtual Circuit. It is a project-defined control abstraction for an admitted forwarding service. The **GVC agent** is proposed userspace software installed on a

participating capable TBox or gateway. It receives instructions from the controller, validates the node-local action, installs standard Linux forwarding state, verifies it and reports status. It is neither a Babel feature nor an MPLS kernel module. Linux forwards the packets after installation; the agent is not a per-packet forwarding loop.

The same physical TBox can host several roles, but the roles retain separate APIs and privileges. A relay normally needs routing, collector and GVC components, not access policy or COSTK. A gateway anchor needs binding, admission, tunnel and NAT capacity. A serving access TBox needs client visibility and may own COSTK sessions. Old proxy sessions may remain on a formerly serving TBox after the client has moved; this is a service endpoint role, not enabling COSTK on a transit backhaul segment.

2.3 Forwarding profiles and permitted fallback

The design distinguishes ordinary routed forwarding, explicit circuit forwarding and anchored mobile service. These are not interchangeable fallback modes. An IP path is usable only if it still reaches the intended service endpoint with the same client identity, proxy owner and security context.

Profile	Selected behavior	Failure disposition
v1 EasyMesh-only	Existing qualified radio-managed forwarding where routed coexistence is unavailable.	Use the vendor-supported recovery process; do not advertise Babel/GVC capabilities.
v2 SOFT/IP	Service-aware IPv4 routing follows the qualified Babel/regional FIB; service tunnels remain explicit endpoints.	Routing reconverges subject to actual link availability; established proxy ownership remains unchanged.
EXPLICIT/GVC	Controller installs a directional hop-by-hop MPLS LSP and production ingress refers to a committed generation.	Use a validated repair or explicitly move eligible ingress to service-preserving IP; unknown transit labels drop.
Fixed CPE / NAT access	One subscriber address can represent a household NAT router.	No promise of individually identified phone mobility or per-phone COSTK policy behind that NAT.
Managed mobile access	Each device has a stable address, authenticated access binding and anchor-delivered traffic.	Retain address/owner on a valid move; report interruptions or failed flows honestly when the anchor or proxy state is lost.

Admission can request best effort, an explicit path, a protected path, or a bounded bandwidth class. A label does not reserve spectrum. If the capacity ledger or protection backend cannot satisfy the request, the controller returns a structured rejection or a separately authorized downgrade; it must not silently label best-effort service as guaranteed.

2.4 Mobility, transport and congestion control are separate

MPLS selects transport next hops. A tunnel identifies a remote delivery or processing context. A stable-address service keeps the client's IP and gateway configuration valid. TCP endpoints maintain the sequence, acknowledgment and congestion-control state. These are four different responsibilities. Encapsulating IP/TCP in MPLS, with or without an outer IP/UDP tunnel, does not by itself authenticate a new attachment, preserve NAT state or move a proxy's sockets.

UDP encapsulation is therefore not required for mobility in this design. Its absence also does not prevent mobility. What matters is that the inner endpoint identity remains stable and the network updates its delivery mapping consistently. The selected standard IP-in-IP service tunnels do not claim transport-layer migration features; they carry the original packet to the correct anchor or proxy context. Protocol-4 reachability and the MTU must be qualified on every participating path. IP-in-IP provides no encryption or peer authentication by itself. [S15]

The initial COSTK policy uses access-link evidence only. A good access link can select bypass for a new flow; a qualifying weak or error-prone access link can select COSTK with the configured client-facing congestion-control algorithm. Hysteresis means different entry/exit conditions and a minimum stable interval, so short metric fluctuations do not repeatedly change the new-flow policy. Existing flows retain their original choice. The exact thresholds are calibrated by the paired test plan, not inferred from RSSI alone.

2.5 EasyMesh v1 coexistence and geographic routing

The source implementation plan's two overlapping EasyMesh zones, controller positions and shared middle row are vendor qualification scenarios, not the selected scalability guarantee. Rows 1–3 and rows 3–5 each contain 15 physical TBoxes; together they represent 30 memberships over 25 boxes, with five shared middle-row boxes. That arrangement cannot assume that the same Agent or radio/MLD accepts commands from two independent controllers. A claimed fifteen-agent scale is not automatically a tested operating limit. Sections 2.6–2.10 define the qualification and safer non-overlapping ownership alternative.

The geographic/DHT-style forwarding proposal is not selected for the initial release. Geographic metadata remains useful for planning, failure-domain grouping and operator display. It is not sufficient to know which radio links currently work, which alternate path avoids interference, or where a moving device is attached. Babel and the PCE use measured topology; the mobility authority uses authenticated bindings. An address allocation number is not a continuously updated physical location.

2.6 EasyMesh scalability of the 25-TBox grid

The physical network remains the 5×5 grid in the supplied drawing. Twenty-five multi-radio TBoxes, their clients, concurrent backhaul adjacencies and controller recovery workload must be supported together. Successfully onboarding 25 idle devices is not sufficient evidence. The limiting resource might instead be controller memory, per-radio client/BSS state, metric processing, association bursts, backhaul depth or contention on a shared radio.

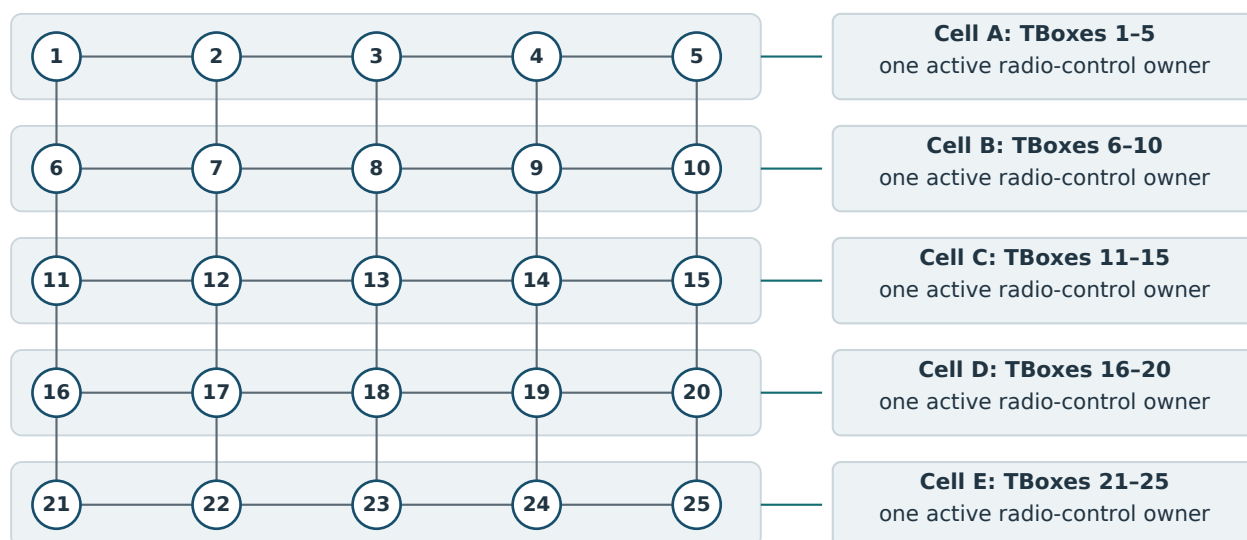
Babel can select routed paths over usable adjacencies, and GVC can install explicit paths. Neither increases an EasyMesh controller's supported Agent capacity, reduces its retained client database automatically, creates independent bAP/bSTA links, or removes RF contention. If the same controller still manages all 25 TBoxes, it still has that management workload after MPLS is enabled. The design must address control scaling and forwarding scaling separately.

EasyMesh implementations expose distinct Agent and Controller roles. For example, prplMesh documents these build roles and separately configurable controller tasks and client state. That supports examining several resource dimensions; it does not establish a capacity number for the target MaxLinear/vendor image. [S17] [S18] This document records a scalability risk and mitigation, not a newly measured root cause on a live TBox.

2.7 Physical grid versus radio-management cells

The preferred scalable architecture is a routed 25-TBox grid that can contain several **bounded EasyMesh control cells**. A cell is an operational radio-management boundary, not a new physical grid, a subscriber IP subnet or an MPLS circuit. Each Agent has one assigned active EasyMesh controller in the selected baseline. The same physical radio/MLD does not acquire two independent configuration owners. Multiple cells can still participate in one Babel grid domain through qualified routed links.

Same physical 5 × 5 TBox grid; example non-overlapping control cells



Lines show intended grid adjacencies, not proven active radio links.

Cross-cell L3 links and discovery isolation require vendor qualification.

Figure 2. Logical radio-management partition of the existing 25 TBoxes. Controller boxes denote roles, not five extra required devices. Cell size and membership remain subject to measured capacity and topology.

The illustrated five cells contain TBoxes 1-5, 6-10, 11-15, 16-20 and 21-25. Five is an example partition size, not an EasyMesh limit or a mandated final layout. It preserves all 25 physical nodes and the intended horizontal/vertical connectivity. Choose actual cell boundaries around verified control reachability, RF relationships, redundancy and resource budgets; another non-overlapping partition may be better. A controller role can be

co-located on a qualified TBox or placed on a supported management device; the drawing does not add five required physical routers.

For the illustrated orthogonal grid, inventory all 40 intended undirected adjacencies: 20 horizontal links within the example row cells and 20 vertical links crossing cells. Actual RF links may differ or add diagonals; record them explicitly. Row-shaped cells are only a simple partition illustration. A failed middle TBox can disconnect a row's EasyMesh control path even while Babel finds a payload detour through another cell. Production cell shapes and control transport therefore need their own redundancy or an explicitly accepted degraded behavior. Do not merge discovery domains to conceal this failure; E03/E07 verify control reachability independently of payload recovery.

Option for the same physical grid	Qualification needed	Design disposition
One EasyMesh domain for all 25 TBoxes	Vendor support and loaded tests for every Agent/radio/BSS/client/hop dimension, recovery bursts and resource headroom	Acceptable when the complete envelope passes; not assumed merely because 25 nodes appear in a topology view.
Several non-overlapping EasyMesh cells	Single-owner radio control, supported local control transport, and working routed cross-cell links	Preferred expansion strategy when one domain cannot safely cover the grid; preserves the physical grid, but requires firmware/interface proof.
Two overlapping 15-node zones	Vendor-defined independent Agent/radio ownership, protected configuration transport and measured coexistence	Experimental source-plan scenario only; shared row-3 membership is not the default redundancy or scaling mechanism.

Dual relay exits provide transport redundancy. They do not require dual EasyMesh ownership of the same Agent. Similarly, a backup controller is not a second active writer: configuration, security and takeover behavior need a supported single-writer failover mechanism. A per-cell controller outage may preserve some established forwarding or may disrupt it; only measured firmware behavior can determine that service claim.

2.8 Qualified cell envelope and admission

Every cell has a versioned scale manifest tied to exact controller/Agent firmware, hardware and enabled feature set. Record vendor-supported maximum, proposed deployment size, largest measured passing load, accepted production limit and reserved headroom separately. Unknown vendor limits remain unknown. A small laboratory passing result authorizes only that tested profile, not extrapolation to 25 nodes or an entire region.

Dimension	What must be counted	Gate
Agent membership	Unique EasyMesh Agent identities; include a controller's co-located Agent and any leaf/relay that actually enrolls	Planned membership and recovery procedure fit the accepted per-controller envelope.
Radio and policy state	Radios/MLDs, BSSIDs, SSIDs, VLAN policies and simultaneous feature combinations	Hardware, firmware and controller tables all pass; multiple radios are not assumed independent control domains.

Dimension	What must be counted	Gate
Client state	Associated and retained client records, per-radio limits, concurrent onboarding/roaming	State and event processing remain bounded at the intended peak, not only with idle APs.
Topology	Usable neighbor degree, controller-to-Agent control hops, payload hops and shared airtime	Both control delivery and application service satisfy measured budgets.
Resources and recovery	Controller/Agent CPU, RSS, flash, requests/s, queue depth, restart burst and recovery time	Accepted headroom remains during loaded recovery; no OOM, watchdog reset or silent loss of authoritative events.

Count an additional household router as an EasyMesh Agent only when it actually belongs to that cell. A standalone AP that merely reports telemetry does not consume an Agent slot, although its reports and clients still consume TBox resources. The immediate scalability gate is the 25 grid TBoxes; downstream devices add workload and must not be hidden from the final capacity plan.

For a homogeneous Agent-count-only planning estimate, $\text{cell_count} \geq \text{ceil}(\text{total_agents} / \text{accepted_agents_per_cell})$. This is a lower bound, not a complete partitioning algorithm: RF connectivity, client load, features and controller reachability can require more cells. Hypothetically, an accepted limit of 12 Agents gives a lower bound of three cells for 25 grid Agents; five five-node cells may still be preferable for topology or headroom. Neither 12 nor five is asserted as a measured target capability. If 50 additional leaf Agents enroll, the total becomes 75 and that same hypothetical lower bound becomes seven. A five-cell diagram must not be used to ignore those extra members.

The deployment service rejects additional enrollment or a feature change that exceeds any accepted dimension. The first response is to stop new management admission and alarm, not evict working Agents or automatically repartition a live grid. Client admission and overload behavior likewise need a supported AP implementation; if the firmware lacks an enforcement interface, restrict deployment administratively and do not claim a software-enforced limit.

2.9 Cross-cell transport and controller coordination

Within a cell, use the vendor-supported EasyMesh controller/Agent discovery and control arrangement. Across cells, use explicit routed infrastructure links. Do not extend a single region-wide subscriber or IEEE 1905 discovery domain to make all controllers appear adjacent. Merely assigning different cell names or VLAN IDs does not prove isolation; interface membership, discovery scope and actual captures must demonstrate it.

Cross-cell links are the essential implementation gate. Both endpoints must support a routed connection without forcing one endpoint's Agent to enroll into the other's EasyMesh domain. Possible backends include a supported independent backhaul interface or a separate Ethernet/point-to-point routed connection. The exact radio manager still owns regulatory and link configuration. A common channel and security policy can be coordinated through the cell owners, but two controllers must not write competing configurations to one radio. If the vendor's bAP/bSTA design ties every neighbor to one indivisible EasyMesh domain, a

cell split cannot be achieved by Babel configuration alone. Obtain a supported firmware/interface change or retain the proven limited configuration; large-grid deployment remains blocked.

Proposed regional management distributes versioned policy and inventory to each cell owner through authenticated northbound APIs. It is custom integration work, not an assumed standard EasyMesh controller-federation feature. Cell controllers retain local supported radio operations. GVC/mobility services keep their existing responsibilities and do not become additional EasyMesh radio writers. Local automatic steering and proposed mobility recommendations must use one configured access-control path, with conflicting independent steering disabled or serialized.

Regional radio summaries are rate-limited and coalesced. Authenticated attachment changes are different: deliver them with event sequence, acknowledgment and bounded replay or an authoritative resynchronization. A dropped RSSI sample may become UNKNOWN; a dropped attachment event must not be replaced by a guessed location. Loss of the regional policy service should not stop already qualified cell-local management. Cross-cell handovers needing an unavailable mobility authority retain the outage semantics in section 9, rather than a new promise of autonomous seamless roaming.

2.10 Cell changes, mobility and fallback

A radio-management cell boundary is not automatically an 802.11r mobility-domain boundary. Cross-cell clients require the same stable-address/anchor design, credential compatibility and qualified FT/key-holder transport described in sections 8-10. Matching SSIDs alone does not establish it. Retain the original COSTK owner after a cross-cell move; changing an EasyMesh controller does not transfer TCP sockets or permit a bypass transition for an existing flow.

A planned split or Agent migration reserves target capacity, stages supported configuration, fences old ownership, onboards and verifies the new owner, restores routed links and reconciles actual attachments before completion. Record expected client disruption and a tested recovery path. The initial procedure uses an approved maintenance window; invisible live repartitioning is not promised. Preserve required old COSTK processes and service paths until their sessions safely drain, even when their old EasyMesh role changes.

If the full 25-node single-domain envelope fails, smaller qualified cells are an alternative only after the cross-cell gate passes. If neither can meet the requirement, the design explicitly requires vendor enhancement or a separately approved replacement radio-management backend. MPLS installation is not an escape from that blocker. The network may retain an already validated smaller service, but cannot be released as a fully scalable 5×5 grid by assertion.

3. Addressing, identity and control-state invariants

3.1 Authoritative IPAM plan

Use one allocation authority before enrolling nodes. The example plan below reserves allocation domains $T = 1$ through 253. T is an IPAM allocation identifier; it need not be the literal number of a town. A grid identifier and geographic metadata are separate inventory fields. If the deployment needs more domains or mobile pools than this plan permits, approve a new non-overlapping plan instead of reusing a reserved range.

Allocation	Example and rule	Ownership
Fixed CPE access	$10.T.N.0/24$, $N = 1-25$ for the illustrated grid. $.1$ is the local gateway; $.2-.9$ infrastructure; $.10-.245$ CPE leases; $.246-.254$ reserve.	IPAM reserves the prefix; the fixed-service lease authority assigns CPE addresses.
Routed links	$10.T.254.0/24$, subdivided into registered $/31$ links or supported $/30$ links. No overlap with service endpoint blocks.	IPAM allocates actual endpoint addresses to named adjacency IDs.
Node loopbacks	$10.T.255.0/24$, unique $/32$ s assigned to nodes.	Inventory binds addresses to node identity, not to a replaceable interface name.
Service endpoints	$10.T.253.0/24$, unique $/32$ outer tunnel addresses; T -specific capacity is explicit.	IPAM assigns endpoint/context pairs. Larger pools require a documented additional allocation.
Proxy server-leg sources	$10.T.252.0/24$, unique $/32$ s assigned only inside the named owner proxy namespace.	IPAM distinguishes these inner socket addresses from root-local outer tunnel addresses.
Local namespace links	$10.T.251.0/24$, registered $/31$ or $/30$ veth subnets, not subscriber leases.	Platform manifest binds each subnet to named namespace/context interfaces and checks capacity.
Stable mobile pools	$10.254.0.0/16$, subdivided by anchor; example $10.254.16.0/20$.	Home anchor owns authoritative leases and aggregated service reachability.
Global control reserve	$10.255.0.0/16$, unused until explicitly allocated.	Central IPAM; never automatically allocated as a town/grid pool.

The fixed node range deliberately stops at 25 in this example; 251–255 are not subscriber node numbers. Other grid sizes require a new allocation rule before deployment. This plan supersedes competing loopback examples such as $172.16.T.N/32$ for new enrollments; it does not automatically renumber an existing network. Migration inventory must identify legacy addresses and provide a separate cutover plan.

An owner server-leg source must not also be configured as a root-namespace local loopback or tunnel address. Otherwise local-address lookup can capture a reply before it reaches the

proxy namespace. The anchor maps its registered alias to `SERVER_IN`; the owner's trusted context routes the decapsulated inner /32 through the correct veth into that proxy namespace. Verify the complete root-to-service route and reapply policy from the receiving interface, rather than assuming a mark or route crosses namespaces correctly.

An anchor /20 gives a bounded mobile allocation pool, not a guarantee that the gateway can support every address as an active device or every device's flows. Sixteen /20 blocks fit in the reserved /16 before exclusions; further anchors require approved pool expansion or smaller blocks. Capacity limits for leases, associations, source-validation entries, host bindings, tunnel endpoints, conntrack and proxy sockets are separate and must all pass.

The original 236-address subscriber example counts CPE/household leases in .10-.245. It is not a claim that a bridged per-device service supports the same number of households with an arbitrary number of phones. Fixed NAT access and managed mobile access have different address, privacy and state-size budgets.

Enrollment rejects an address that overlaps a fixed subscriber prefix, mobile prefix, node loopback, routed link, service endpoint or active legacy allocation. Address uniqueness is tested both in IPAM and on the actual segment. A mobile example is 10.254.16.23 with gateway 10.254.16.1 and client mask /20. Its network binding uses a /32 host route; these are different uses of a prefix length.

3.2 Identity model and state ownership

Every node has a stable provisioned `node_id`, current `boot_id`, firmware identity and certificate identity. Interface names and ifindexes are observed properties, not durable identities: an interface may be recreated during a radio restart. A `link_id` binds the peer node, local/remote radio or MLD, logical interface and current incarnation. A MAC address alone is not accepted as proof of a trusted node.

A managed mobile has an enrollment or service identity, an address lease, and its current authenticated attachment. A private/randomized station MAC can identify a current association but is not assumed stable forever. Re-enrollment and station-MAC changes require a policy that binds the new association to the legitimate service identity; they must not let a different device claim an existing lease or flow. Per-device mobility is unsupported where the access system cannot establish this mapping.

State object	Authority	Lifetime and invariant
Prefix/endpoint allocation	IPAM	Durable until controlled retirement; no overlap or automatic reuse.
Link observation	Collector at the reporting node	Boot-scoped sequence and freshness; stale observations cannot advertise a new usable path.
Circuit generation	GVC leader and local agents	Unique generation plus locally allocated labels; retain old forwarding until references and drain conditions permit removal.
Mobile binding	Home anchor / mobility authority	Monotonic binding epoch selects one authorized current delivery edge.

State object	Authority	Lifetime and invariant
TCP mode and owner	Anchor flow-admission authority	Immutable for a flow incarnation; a first FIN is not permission to discard ownership.
TCP sockets	Owning COSTK process or original endpoints	Process-generation-specific; metadata replication does not recreate lost TCP state.

The controller's leadership term, circuit generation, mobile binding epoch, AP association sequence and process generation are different counters. They must not be collapsed into one ambiguous version field. A client can roam without changing a circuit; a circuit can be repaired without roaming; a COSTK process can restart while the address lease remains valid. The APIs carry the relevant identifiers explicitly.

3.3 Control security and no-feedback enforcement

The control network permits only enrolled nodes to call authenticated, versioned management APIs. Use mutually authenticated transport with role authorization, bounded request sizes, expiry and replay protection. Certificate issuance and revocation integrate with the platform's actual secure onboarding system; the design does not require a new independent root of trust on the constrained leaf.

One fenced leader owns each mutable control object. Nodes persist the highest accepted leadership term or obtain an equally strong fencing decision before applying a write. A request includes object ID, desired generation, expected previous generation, operation ID, expiry and a canonical content hash. Retries with the same operation ID and payload are idempotent; conflicting reuse is rejected and audited. An expired leader cannot overwrite a newer binding or label generation.

Separate unprivileged protocol parsing from narrowly scoped privileged forwarding operations. A GVC helper may manipulate only owned labels, tables, tunnel endpoints and interface settings from its manifest. COSTK has no credentials for that helper or the Babel metric interface. Its lifecycle publisher can state that an admitted owner process exists or that a flow has safely closed; it cannot request a path, change a metric or switch a binding. Tests must attempt forbidden writes, not merely inspect a diagram showing separate components.

The selected kernel data plane also needs a trust boundary. MPLS ingress is enabled only on approved transit interfaces; subscriber ports cannot inject labels. IP-in-IP packets are accepted only from expected outer endpoint pairs on the permitted routed infrastructure interfaces. These filters prevent accidental or unauthorized context selection but do not make unencrypted IPIP safe on an untrusted transport. If the backbone is outside the trust boundary, add and qualify an authenticated encryption transport and recalculate MTU, CPU and failover budgets before release.

3.4 Persistence, loss and release invariants

Persist desired control state and operation outcomes with a crash-consistent store. Do not write high-rate telemetry samples or every packet counter to flash. Local agents reconstruct observed kernel state after restart, compare it with owned desired state and reconcile conservatively. A resource whose ownership is uncertain is quarantined from reuse until the controller proves it can be retired.

Mobile leases and committed bindings must survive a control-process restart. Surviving proxy processes can retain their sockets while control reconciles. A proxy process restart, gateway NAT-state loss or loss of a non-replicated anchor cannot be disguised as transparent continuity. The system reports affected flows, blocks unsafe reclassification of their established packets and allows explicitly authorized new connections. High availability for metadata and high availability for TCP/NAT state are different engineering deliverables.

These invariants are release requirements: no second owner for the same live flow incarnation; no delivery to an unauthenticated attachment; no stale controller overwriting a newer generation; no arbitrary IP fallback for an unknown MPLS label; no repeated proxy interception of an already classified service leg; and no flow-mode change merely because RSSI crossed a threshold. Later performance optimizations must preserve these invariants rather than silently weaken them.

4. Routed underlay and multi-link telemetry

4.1 Routing responsibilities and boundaries

Each grid is an independently operable routed domain. Babel provides reachability between its TBox and Relay locators; eBGP connects explicitly configured grid, regional and gateway routing domains. EasyMesh and the radio driver establish usable links. The mobility controller determines the current delivery attachment of a subscriber. These are separate decisions: changing a radio channel, selecting a routed next hop and moving a subscriber binding must not be treated as the same event.

A physical 5×5 grid may contain several bounded EasyMesh radio-management cells under sections 2.6–2.10. These cells need not be separate Babel domains. A cell boundary is not automatically an eBGP boundary or a new address allocation. The underlay sees only the cross-cell adjacencies whose vendor-supported establishment and control isolation actually passed qualification.

Babel is a distance-vector protocol with loop-avoidance mechanisms, not a complete topology-distribution protocol. Its neighbor and route records are useful inputs, but do not automatically reveal every link, shared radio or failure domain. Its standard packets do not distribute our circuit identifiers or MPLS bindings. We use the deployed Babel implementation without custom GVC protocol extensions. [S1]

Phase 1 starts with configured, stable Babel link costs and explicit prefix filters. Each TBox originates its assigned locator and infrastructure prefixes only. Relays export an allowlist of authorized grid prefixes and import approved infrastructure routes and qualified exits. Default-route origination requires gateway-health policy, not merely an established BGP session. Redistribution rules must reject unauthorized subscriber, management and overlapping prefixes and prevent routes from being reflected indefinitely between Babel and eBGP.

Subscriber addressing and infrastructure addressing have different owners. A TBox must not start originating an entire mobile address pool simply because one mobile attaches there. The mobility chapter defines subscriber delivery state. Gateway selection can respond to exit health for a fixed/non-mobile service or initial home-anchor assignment. An enrolled managed mobile still sends both existing and new flows through its assigned home anchor; an underlay exit change is not an anchor change. A live NAT-anchored or COSTK-owned connection cannot adopt an unrelated gateway merely because it has become the shortest route.

Multiple north, south, east and west links must remain separately identifiable where the hardware exposes them. Each adjacency records its actual interface, peer, radio and channel. If firmware exposes one opaque bridge or one combined radio interface, the system must advertise that limitation; neither Babel nor the GVC controller may invent independent links or protection diversity behind it. Underlay management remains reachable without relying on the circuits that it controls.

4.2 Collector lifecycle

The proposed topology collector builds a directed engineering graph from authenticated node inventory, driver observations, Babel state and bounded probes. It runs centrally, with small local adapters on capable TBoxes. The leaf mesh reporter supplies client access facts only; it does not become a topology controller. This collector is new software, not functionality already delivered by Babel.

The processing lifecycle is Discover → Validate → Measure → Stabilize → Advertise. Discover creates a provisional record for a seen peer/interface. Validate confirms node identity, inventory, permitted role and interface incarnation. Measure attaches driver counters and probe samples with their definitions. Stabilize rejects stale/reset samples and applies the configured smoothing policy. Advertise publishes a versioned graph snapshot and events for material changes. Withdrawal or expiry marks an edge unusable for new constrained paths while preserving the historical reason.

Record group	Required fields	Interpretation
Identity	node_id, boot_id, cell_id, radio_controller_id, interface_id, interface_incarnation, peer_id, peer_interface	Distinguishes management ownership, replacement hardware, reboot and interface recreation.
Link	direction, carrier, association_state, auth_state, radio_id, channel, width, risk_groups	Radio and power dependencies are explicit, not inferred from geography alone.
Observation	sample_sequence, local_monotonic_time, received_at, window_ms, validity_bitmap	Identifies freshness, interval and supported measurements.
Quality	rss_i_dbm, tx/rx_rate_bps, retry_count, failure_count, airtime_busy, probe_rtt_ms, delivered_bps	Unknown values remain unknown; raw counters retain their units and provenance.

Driver adapters also record counter width and reset behavior. Rates are computed from valid deltas over an observation interval; a reboot or wrap ambiguity invalidates that interval. Retry attempts, failed frames and lost IP packets are not interchangeable denominators. PHY rate is not available application bandwidth. Unsynchronized clocks cannot provide reliable one-way delay, so the baseline uses round-trip probes and local queue observations where supported.

An interface can be administratively present but unable to carry the intended peer traffic. Therefore, link eligibility combines authentication, driver association, Babel neighbor health and an appropriate data-path probe. It does not depend on RSSI alone. Evidence is directional: good reception from B at A does not establish equal quality from A to B. The collector maintains both directed edges and their confidence levels.

Every numeric observation has VALID, UNKNOWN, STALE or INVALID status. A missing driver retry counter is not zero errors. A candidate lacking capacity evidence can carry best-effort traffic under policy, but cannot satisfy a minimum-rate admission constraint. Retained historical measurements may assist diagnosis; they do not silently renew eligibility.

4.3 Damping, publication and consumers

For each supported metric, the adapter calculates a windowed sample and an exponentially smoothed value. The smoothing interval, maximum sample age, minimum sample count and change threshold belong to a named policy version. Initial laboratory settings can use one-second summaries, five-second freshness and a multi-second improvement hold; these are commissioning parameters, not service guarantees. Hard association or authenticated peer loss immediately invalidates the affected adjacency, without waiting for the improvement hold.

The collector publishes full snapshots with `graph_revision` and incremental events with an ordered sequence. Consumers detecting a sequence gap obtain a new snapshot. A publication contains the source boot/interface incarnations and measurement policy version so a controller can explain the exact inputs used to select a path.

The GVC path engine reads this graph. A later Babel metric adapter may read independently qualified raw link-quality measurements through a supported local Babel interface. It must not rewrite Babel's feasibility state, and the fixed-cost baseline remains available if the adapter is unsupported. GVC utilization, reservation success and COSTK throughput never feed back into Babel metrics. COSTK reads access quality and delivery/owner facts; it does not adjust routes or circuit costs.

Phase 1 tests must show accurate discovery and withdrawal, asymmetric links, counter resets, stale telemetry, neighbor flapping, redundant exits and bounded publication load. Phase 2 replays those recordings into the path engine. The important result is reproducible decisions from recorded evidence, not an attractive live topology display.

5. Grid Virtual Circuit control and MPLS forwarding

5.1 What the GVC agent is

GVC means Grid Virtual Circuit: our management abstraction for an aggregate, unidirectional delivery contract. The proposed gvc-agent is a userspace daemon on forwarding TBoxes and corresponding gateway endpoints. It is not a kernel module, a new Babel protocol, or software for the memory-limited leaf mesh router. Its controller chooses paths; the agent validates and installs its node's portion; the Linux kernel forwards packets. These components remain implementation work described by this document.

The agent exposes capability discovery, resource allocation, prepare, verify, activate, query, abort and retire operations. It restricts writes to allocated label ranges, approved IP policy tables, trusted interfaces and qualified queue/accounting objects. It cannot modify Babel/eBGP configuration, run arbitrary remote shell commands or enable subscriber-facing MPLS input. A privileged helper can isolate kernel writes from RPC parsing; memory, message size and concurrent transactions are bounded.

The initial service modes are deliberately precise. SOFT uses ordinary, service-aware IP forwarding and supplies classification, accounting and policy metadata. It does not install a static MPLS route and pretend that route follows later Babel changes. EXPLICIT installs a controller-selected hop-by-hop MPLS path. PROTECTED is a protection attribute of an explicit circuit after its detection and repair mechanism passes Phase 4; it is not automatically supplied by MPLS or ECMP.

An IP fallback is eligible only if its complete path preserves the selected gateway/NAT anchor, subscriber service and original TCP owner. If the underlay contains only TBox locator routes and cannot deliver the original subscriber destination, removing MPLS does not create a valid fallback. Such a service must retain a qualified delivery mechanism or report an outage.

5.2 Intent, compilation and state ownership

Circuit intent identifies an aggregate service, never an individual TCP session. Admission and forwarding selectors include an authorized ingress, destination/service scope and class. Reverse traffic requires a separate circuit. A shared pair of transport circuits can serve many subscriber bindings and many connections while edge flow ownership remains separate.

```
CircuitIntent:
  circuit_uuid, tenant_id, service_id, policy_version
  ingress_node, egress_node, direction=UNIDIRECTIONAL
  mode=SOFT|EXPLICIT, protection=NONE|LOCAL_REPAIR
  selector={ingress_scope, destination_scope, traffic_class}
  constraints={min_rate_bps?, max_probe_delay_ms?, max_hops?}
  diversity={link|internal_node|shared_risk}
  fallback=QUALIFIED_IP|DROP
```

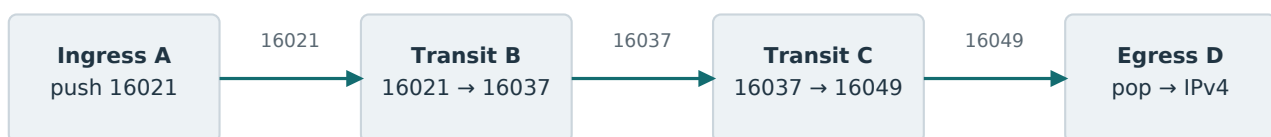
```
CircuitGeneration:
  circuit_uuid, generation, controller_epoch, transaction_id
  expected_previous_generation, graph_revision, config_hash
  primary_adjacencies[], repair_paths[], local_bindings[]
  resource_holds[], verification_results[], state, reason
```

The controller owns intent and graph-based path selection. Each agent owns its local label allocator and returns reservation tokens; the controller compiles upstream outgoing labels from the next node's accepted incoming allocation. This prevents two controller transactions from independently assigning the same local label. Label ownership is scoped to a network namespace and agent boot incarnation, with a persistent journal sufficient for reconciliation.

Generation numbers identify management updates; controller epochs identify writer authority. Neither is an extra MPLS header field. An agent rejects stale epochs, unexpected previous generations, conflicting reservations, unauthorized interfaces and action sets outside its capability manifest. Repeated requests with the same transaction ID and configuration hash return the recorded outcome; reusing an ID with different content is an error.

5.3 Concrete native-MPLS baseline

The production starting point is controller-installed hop-by-hop label switching, not Segment Routing. At ingress, an authorized IPv4 FEC receives one transport label. Each transit node swaps that label for the value allocated by the next node and forwards on the specified adjacency. The egress removes the final label and delivers the original IPv4 packet. No UDP header or TCP modification is introduced. [S2] [S3]



Each receiving node allocates its own incoming label; one label is on the wire.

Generation is authenticated control state, not an extra MPLS header field.

Forward and reverse directions are separate LSPs; install downstream first.

Figure 3. Example hop-by-hop LSP. The label numbers are illustrative allocations, not reserved values or an executable configuration.

For example, A sends label 16021 to B; B maps 16021 to label 16037 toward C; C maps 16037 to label 16049 toward gateway D; D terminates 16049. The numbers are illustrative local incoming labels. They are not node IDs, globally unique identifiers or traffic sequence numbers. The opposite direction has independently allocated state.

Standard Linux MPLS uses the top label within a network namespace as its forwarding-table key. It does not match a separate generation field, and its label lookup is not an interface-plus-label table. Consequently, each generation receives distinct incoming labels where coexistence is required. Interface admission is a separate security control. Labels must be unique within their actual platform namespace even when the controller models distinct logical domains. [S5]

The first MPLS forwarding qualification carries one IPv4 service. Its egress backend pops the bottom label to the local loopback delivery path, then performs ordinary IPv4 routing. The lab must prove this specific delivery path, including source checks and the absence of accidental re-encapsulation. This is a concrete backend to qualify, not a claim that every vendor kernel already implements it correctly. Where appropriate, a tested directly connected egress next hop may be used instead.

For the mobility and COSTK service, the selected backend additionally uses standard IPv4-in-IPv4 encapsulation. All client traffic first reaches its home anchor; owner-return traffic goes through that anchor to the current access TBox. Dedicated outer endpoint pairs and verified decapsulation devices distinguish CLIENT_IN, CLIENT_DELIVERY, OWNER_INPUT, SERVER_OUT, SERVER_IN and CLIENT_RETURN directions. Access/anchor tunnels carry the first two; separate anchor/owner endpoint pairs distinguish client-leg input, server-leg traffic and client-leg return. MPLS carries the outer service-endpoint IP packet; ordinary endpoint routing delivers it to the selected IP-in-IP tunnel. This avoids pretending that a local packet mark survives transmission across the grid. The mobility chapter specifies those directional paths. [S15]

The resulting service packet is MPLS transport label | outer IPv4 | inner IPv4 | TCP | data, still without UDP. For one label and an outer IPv4 header without options, overhead is 24 bytes beyond the inner IP packet. The simpler native MPLS | IP | TCP format remains the basic forwarding test; it is not claimed to carry the additional directional context by itself. IP-in-IP is not encryption or peer authentication. Endpoint allowlists, trusted transport and tunnel-demultiplexing tests are mandatory. Phase 6 must prove that decapsulation re-enters the intended firewall/TPROXY hooks; the design does not rely solely on MPLS loopback reinjection to establish proxy service context.

Optional service labels are disabled until a service-context backend is demonstrated. A second label does not automatically select a VRF, tenant, socket owner or routing table. A future namespace backend could terminate a service label onto a dedicated veth delivery path into an isolated service namespace, where its own routes and firewall apply. That is an alternative requiring explicit tests of label-to-namespace mapping, MTU, isolation, counters and COSTK hooks, not an additional Phase-5 dependency. Multiple overlapping tenant address spaces are outside the single-service baseline until their isolation backend passes qualification.

Linux 4.4.60 permits at most two newly imposed labels per relevant operation in upstream MPLS code. This does not mean that every received stack is limited to two labels; it means that the proposed ingress cannot simply impose a long segment list. [S4] [S6] SR-MPLS is therefore a later, separately specified encoding. If adopted, a Binding SID selects a policy at its owning headend and is expanded/consumed there; it must not be left below that same policy's complete segment list as an end-to-end circuit identifier. [S8]

5.4 Transaction and activation protocol

The control RPC envelope contains `api_version`, authenticated principal, bounded operation list, transaction ID, controller epoch, fencing-grant reference, circuit UUID, generation, expected prior state and configuration hash. Agents validate a live grant from the controller authority before accepting mutations. A larger integer supplied by a peer is not itself proof of leadership. Controller HA must provide fenced authority; signed but expired or replayed requests are rejected.

Operation	Agent action	Required result
Reserve / Prepare	Reserve labels and install downstream entries and qualified ancillary objects.	Allocation tokens, normalized installed state and preparation deadline.
Verify	Read back entries and admit only authorized candidate-path probes.	Hash match, probe evidence and capability/resource status.
Activate / QueryState	Replace the permitted ingress selector; report actual selected generation.	Durable activation result or explicit uncertainty requiring reconciliation.
Abort / Retire	Remove unused prepared state or safely unreferenced old state.	Released resources and tombstones preventing premature label reuse.

Preparation proceeds from egress toward ingress. “Prepared” means installed but not referenced by production ingress; a standard Linux MPLS route is not intrinsically inactive. Probe injection must use a separate authorized classifier and cannot enable subscriber traffic onto an incomplete path. Each agent records the normalized result of its successful kernel operations before acknowledging readiness.

The activation point is an ingress route or classifier replacement supported by the chosen backend. The first implementation uses one controlled selector per activation. A batch of netlink writes is not an atomic transaction across tables or nodes. If a service requires several selectors, the implementation must provide a tested common indirection or explicitly serialize/gate those selectors. It may not claim all-or-nothing activation merely because all messages were sent together.

On prepare failure, abort only the new, unreferenced generation. On an activation timeout, query the ingress and required agents before deciding the result; a lost reply does not prove failure. Preserve working forwarding during reconciliation. Old-generation removal requires proof that no ingress or repair action references it, followed by a drain interval covering measured buffering and in-flight traffic. A zero counter alone is not sufficient proof. Unreachable agents keep their labels quarantined from reuse until reconciliation establishes safe retirement.

Preparation holds may expire automatically. Committed forwarding is not silently garbage-collected merely because the controller connection is lost. Agents retain the last authorized state, reject unauthorized changes and follow the explicitly configured service-outage policy. If an administrative forwarding lifetime is required, it must specify safe ingress quiescence or fail-closed behavior separately from resource reclamation. Long-lived COSTK-owner references also prevent retirement of required owner paths.

5.5 Correct miss, failure and restart behavior

An unlabeled packet that does not match an enabled circuit may use its authorized IP service path. In the directional IP-in-IP backend, ingress fallback may retain the same outer tunnel endpoints and use ordinary IP routing to them, provided that route and decapsulation context are qualified. A labeled packet with no LFIB entry is different: it is dropped and reported. Generic label stripping on a transit miss could create a loop or bypass information that exists only in the label binding. The baseline provides no such behavior. [S2] [S5]

After reboot, the agent starts with subscriber MPLS admission disabled, loads its identity and journal, inventories actual kernel state and reconciles with current authority. Stale interface indexes are not reused as identities. It enables an ingress only after downstream verification or a qualified service-aware fallback decision. Babel and management routes remain independently operational. Phase 3 fault injection must cover every operation boundary, duplicate requests, process crashes, lost replies, interface recreation, stale authority and two coexisting generations.

6. Protection, QoS and scaling

6.1 Path computation and repair contracts

The path engine first excludes unauthenticated, unusable, stale or unsupported adjacencies. It also rejects paths that exceed label, queue, MTU or temporary-generation limits. Required rate or delay evidence must exist before the corresponding constraint can be evaluated. A reachable Babel locator helps validate the intended endpoint, but the explicit next hop need not be Babel's currently selected IP next hop; otherwise explicit traffic engineering would have no meaning.

For eligible edges, the prototype uses a nonnegative weighted cost combining normalized link cost, measured delay, loss/retry penalty, damped load and shared-channel/risk penalties. Every weight, normalization bound and policy version is recorded with the decision. The engine runs a constrained shortest-path search and then checks whole-path constraints. Equal-cost candidates prefer the current path, fewer hops and greater measured headroom. An improvement threshold and hold timer suppress optimization churn; a confirmed failure invalidates the path immediately.

The first reproducible implementation enumerates up to a configured K loopless candidates using a shortest-path plus Yen-style candidate enumeration. Start with K = 16 as a laboratory search bound, not a completeness guarantee. Use strictly positive configured edge costs first; enable measured cost terms only after their normalization and missing-data policy pass replay tests. Check additive delay, hop limit, service-context support, MTU and expanded capacity demand for each candidate. For protection, compile candidate local-repair sets for each eligible primary rather than treating an end-to-end backup as a repair at every node. Try alternative sets when the first set fails capacity. Limit candidate sets to R = 16 per primary initially; report search exhaustion explicitly. Lexicographic node/link IDs break any remaining tie, so identical graph and policy snapshots yield identical output.

```
plan(intent, graph_revision, policy_version):
    graph = validate_and_filter(snapshot(graph_revision), intent)
    for primary in bounded_loopless_candidates(graph, K):
        if not path_and_service_constraints_hold(primary):
            continue
        sets = [empty_repair_set]
        if intent.requires_protection:
            sets = bounded_repair_sets(primary, failure_scenarios, K, R)
        for repair_set in sets:
            if not compile_and_verify_repair_set(primary, repair_set):
                continue
            demand = expand_service_and_failure_demand(primary, repair_set)
            if reserve_resources_compare_and_swap(demand):
                return prepare_generation(primary, repair_set, demand)
    return NO_ADMISSIBLE_CANDIDATE_WITHIN_SEARCH_BOUND
```

The search result must distinguish policy rejection, stale/missing evidence, resource exhaustion and bounded-search exhaustion. It must not claim that no physically feasible pair exists merely because the bounded search found none. The resource compare-and-swap rechecks ledger revision and graph freshness immediately before prepare; a concurrent admission or withdrawal restarts planning instead of oversubscribing the old snapshot.

Phase 2 compares small graphs against exhaustive path enumeration and records counterexamples to the bounded heuristic.

Backup computation excludes the protected internal nodes, links or risk groups, preserving the required endpoints. Each repair record names its protection point, protected resource, merge point, ordered bypass and preserved service context. A node-diverse path cannot protect loss of the service endpoint itself. A path through a second Relay may protect Relay transport while still targeting the original gateway anchor; switching to an unrelated NAT gateway is not equivalent repair.

For each declared failure scenario, the compiler identifies the point of local repair and a valid downstream merge point, allocates the distinct repair labels and verifies the complete post-repair walk to the unchanged service endpoint. On primary $A \rightarrow B \rightarrow C \rightarrow D$, an $A \rightarrow E \rightarrow F \rightarrow D$ backup is useful at A, but gives B no action for its B-C failure. B needs its own verified bypass to C or a later merge point, depending on whether the protected failure is a link or node. The generation is PROTECTED only for scenarios whose actual local actions and capacity pass. Initial protection covers one declared link/node/shared-risk event at a time. Simultaneous independent repairs are unsupported unless their composition is explicitly compiled, loop-checked and tested; there is no assumption that individually safe repairs remain safe in combination.

The baseline local-repair mechanism is a userspace action. A local detector reports failure to the agent; the agent validates an already prepared bypass and performs a single active LFIB next-hop replacement. This can operate without the central controller when its repair authority and prepared state remain valid. It still has detection, scheduling and kernel-update latency, which must be measured. Upstream 4.4.60 MPLS ECMP does not supply the desired prioritized backup behavior; a selected down output can cause drops. [S5]

Each activation is tied to the current circuit generation and interface incarnation. A stale detector event cannot overwrite a newer circuit. After repair, the agent reports BACKUP_ACTIVE, the triggering evidence and selected bypass; the controller later restores a valid primary/backup arrangement transactionally. If backup readiness is uncertain, apply qualified fallback or fail closed. Do not retry arbitrary neighboring outputs.

Phase 4 records detection time, local processing time, netlink completion, first successfully delivered packet and later global restoration. A sub-50 ms target may be evaluated when alternate adjacency is already usable, but is not a promise. Hardware protection groups and FRER are optional backends after exact-variant proof. Replication requires duplicate elimination and roughly twice the protected traffic load; otherwise it delivers duplicates rather than hitless protection.

6.2 Admission and scheduling

MPLS path state reserves no radio airtime by itself. The initial admission service is conservative engineering policy, not a hard latency or bandwidth guarantee. Capacity estimates come from measured distributions under realistic contention, not PHY rates or an isolated peak test.

An admission transaction holds capacity on each bottleneck and on each shared airtime domain. Two links using the same radio/channel may consume the same resource even when

they are distinct graph edges. The ledger therefore tracks both directed-link demand and shared-resource demand. Repeated same-channel hops and simultaneous backup activation must not be counted as independent free capacity.

Charge every service traversal, not just the application's final throughput. For a download of X bit/s with both owner and current edge remote from the anchor, the anchor grid side sends approximately $2X$ of payload (SERVER_IN and CLIENT_DELIVERY) and receives X (CLIENT_RETURN), before ACKs, encapsulation and retransmissions. The Internet side separately receives the server payload. Shared links can carry several of these passes. The compiler expands a service graph into directed paths, sums each path's demand on every shared resource, and admits against those totals. Collocated roles or independent links change the actual load and must be represented, not assumed to cancel it.

The baseline uses dedicated backup budget where a protected rate is requested. If a deployment instead shares backup capacity among circuits, it must state the assumed failure scenarios and calculate simultaneous demand under each scenario. Without that evidence, protection guarantees reachability only and may deliver lower throughput after failure. Capacity holds expire on abandoned preparations; committed reservations change only with recorded admission, preemption or retirement transactions.

Classes are CONTROL, REALTIME, PREMIUM and BEST_EFFORT. CONTROL receives a small bounded queue and admission/policing so subscriber-marked traffic cannot starve the node. The remaining classes use configured scheduling weights and bounded queues. DSCP markings from untrusted clients are classified or remarked at the access edge; they are not accepted as authority. The exact qdisc, driver queues and airtime behavior are platform capabilities, not abstract MPLS functions.

Tests saturate primary and backup resources concurrently, inject shared-radio failures, introduce competing classes and restart the ledger during a transaction. Success requires no unexplained double booking, bounded control service and accurate downgrade events. Any claimed committed-rate service remains blocked until physical-grid evidence supports it.

6.3 Resource budgets and observability

Circuits aggregate by ingress/egress, service and class. They do not scale as subscribers multiplied by TCP connections. Nevertheless, a path crossing several nodes consumes several local LFIB entries, and primary, repair and temporary generations multiply that cost. A 25-node grid with four classes has 100 endpoint/class combinations only under a specified endpoint model; that number is not a bound on total LFIB state. All-to-all services require a different budget.

Each agent reports limits and current use for labels, IP routes, selectors, accounting rules, queues, reservations, retained generations, memory and maximum newly imposed labels. The controller reserves repair and make-before-break headroom before admission. Resource exhaustion rejects new low-priority work; it does not evict a referenced live path implicitly. Inter-grid services use separate grid and backbone contracts with explicit stitching so local topology churn need not rebuild a national path.

Mandatory evidence comprises ingress selection counters, egress delivery counters, bounded path probes, interface counters and control-operation results. Upstream 4.4.60

LFIB route dumps do not provide the required per-label packet/byte counters. Transit per-label statistics therefore require a qualified accounting backend, such as a supported tc classifier, with measured rule and CPU costs. Until that backend passes, reports label transit counters unavailable; they must not invent zeros or claim hop-by-hop accounting. [S4] [S5]

Every circuit status includes active generation, path, fallback eligibility, repair state, last verification, resource use, owner references and telemetry age. The operator trace command distinguishes installed state from observed delivery. Phase 7 measures the combined overhead of routing, telemetry, GVC, mobility and COSTK; isolated daemon RSS is not a complete router memory budget.

6.4 EasyMesh control load and regional scaling

Partitioning the 25-TBox grid's radio-management workload bounds local control scope; it does not remove controller work. Keep separate ledgers for EasyMesh Agent/radio/client state, GVC forwarding entries, mobile bindings, proxy sockets and anchor traffic. A larger controller can relieve CPU or memory pressure but cannot override an Agent's hard limit, an unsupported radio combination or an unformed backhaul link.

Measure steady report volume and bursts separately. A first-order planning model is $\text{reports_per_second} = A/T_{\text{agent}} + R/T_{\text{radio}} + S/T_{\text{client}} + \text{event_rate}$, where A, R and S are counted objects and T values are actual reporting intervals. This is not a protocol-complexity guarantee: batching, multipart messages, request/reply traffic, retransmissions and controller-specific polling alter wire and CPU load. Capture bytes and processing cost, then charge traffic on every traversed backhaul hop. Raw per-client polling across all grids is not the regional design.

The proposed cell adapter uses bounded subscriptions/batches, local raw samples, regional summaries, rate-limited diagnostics and an explicit cap on candidate-AP queries. Full diagnostic scans are on-demand and scheduled so that all 25 nodes do not rescan simultaneously. Prevent duplicate polling by COSTK, GVC and management processes through the existing collector interface where supported. Preserve required event reliability and resynchronization; never rate-limit away authorization while continuing to claim an authoritative client binding.

Regional growth adds accepted cells with independently bounded local workloads. It still increases aggregate northbound traffic, topology state, mobility events and anchor capacity, so regional services need their own measured limits and partitioned work queues. A regional outage must not make every cell restart or rediscover simultaneously. Reconnect uses jitter, bounded replay and state snapshots. Hard local controller limits stop further enrollment; sustained resource pressure triggers a capacity alarm and a planned split/upgrade, not an automatic controller change during active sessions.

No fixed cell size or statistical compression ratio is claimed to solve all grid scales. E01-E10 measure the actual controller/Agent workload, isolation and recovery behavior. The complete grid is released only if every cell, every required cross-cell link and the shared regional/anchor services meet the accepted envelope together.

7. Platform and kernel implementation

7.1 Captured platform and required rebuild

The captured Edge TBox configuration in `build/edge-4.4.60/kernel.config` enables MPLS framing and modular MPLS GSO, but disables MPLS routing and the lightweight-tunnel framework. It is evidence about that captured image, not proof that every future Grid or Relay TBox uses the same firmware.

```
CONFIG_MPLS=y
CONFIG_NET_MPLS_GSO=m
# CONFIG_MPLS_ROUTING is not set
# CONFIG_LWTUNNEL is not set
CONFIG_NET_IP_TUNNEL=m
CONFIG_PROC_SYSCTL=y
```

For that image, complete IP-to-MPLS forwarding requires a firmware rebuild using the exact vendor tree, configuration and cross-toolchain. `CONFIG_LWTUNNEL` is built-in infrastructure; copying a routing module cannot enable infrastructure compiled out of the kernel. Upstream MPLS IP tunnel support depends on both `LWTUNNEL` and `MPLS_ROUTING`. [S7]

Configuration	Baseline setting	Purpose
<code>CONFIG_MPLS /</code> <code>CONFIG_LWTUNNEL</code>	y / y	MPLS and route-encapsulation infrastructure.
<code>CONFIG_MPLS_ROUTING</code>	m or y	Native label receive and forwarding.
<code>CONFIG_MPLS_IPTUNNEL</code>	m or y	IPv4/IPv6 route-based label imposition.
<code>CONFIG_NET_MPLS_GSO /</code> <code>CONFIG_PROC_SYSCTL</code>	m or y / y	Correct GSO handling and platform/interface controls.

Expected modular outputs are `mpls_router.ko`, `mpls_ip_tunnel.ko` and `mpls_gso.ko`. Existing `NET_IP_TUNNEL` support supplies generic tunnel infrastructure; it is not a substitute for MPLS label imposition or the specific IP-in-IP driver. Matching a version string is insufficient: architecture, vendor patches, symbol versions, configuration and package ABI must match. Never force-install a foreign kernel package.

The same captured configuration also disables `CONFIG_NET_IPIP`, `CONFIG_NETFILTER_NETLINK_QUEUE`, `CONFIG_NETFILTER_XT_TARGET_NFQUEUE` and `CONFIG_NETFILTER_XT_TARGET_TPROXY`. Phase 5 must include matching IP-in-IP driver support for the selected directional tunnel backend. Phase 6 must qualify matching `NFQUEUE` and `TPROXY` support, their configuration dependencies, userspace bindings and firewall rules for controlled admission and interception. The live target may differ from the captured image; runtime evidence must decide what is actually absent. These are standard-kernel integration/build gates, not authority to force incompatible modules into the device.

The selected service-isolation prototype additionally requires network namespaces, veth interfaces and policy routing. The saved configuration has `CONFIG_NET_NS=y`, `CONFIG_VETH=m` and `CONFIG_IP_MULTIPLE_TABLES=y`; verify the matching veth module, namespace construction and

actual packet path at runtime. This requirement is distinct from the deferred overlapping-tenant/service-label backend. The saved `CONFIG_NETFILTER_XT_MATCH_SOCKET` is unset; enable and qualify it if the interception rules use socket matching. Root-to-service veth routing, listener SYN-ACK routing and server-leg tuple isolation are explicit Phase 5/6 tests, not consequences of an MPLS build passing.

7.2 Installation and capability manifest

Phase 0 records release, kernel version, architecture, configuration hash, module vermagic, toolchain/vendor identity, relevant device types and firmware recovery method. It tests whether installed `ip` supports the necessary operations and adds the matching full utility package only when required. Package names and package-manager commands must follow the actual OpenWrt release; a current feed is not evidence that a package exists for the legacy image.

The integration package loads the selected modules and configures a bounded platform-label table. MPLS input is enabled only on allowlisted backhaul interfaces after `netifd` creates them. Leaf/client ingress remains disabled and is independently filtered where required. Hotplug handlers verify interface identity before applying policy. Reboot tests must cover cold boot and interface recreation, not only manual module insertion.

The capability manifest distinguishes supported, unsupported and untested operations. It records newly imposed label depth, pop/reinjection behavior, MTU/GSO behavior, relevant firewall/TProxy hooks, accounting backend, local-repair method and any offload limitations. Software capability does not imply line-rate forwarding. LAN969x offload requires the exact silicon variant and driver evidence; unsupported offload must not divert packets around required service processing.

7.3 Packet-size and forwarding qualification

Each native MPLS label adds four bytes. The effective subscriber MTU must fit the smallest path payload budget including every possible primary and repair encapsulation, including the selected IP-in-IP service header. Linux 4.4.60's baseline MPLS paths can drop oversized packets without generating the desired end-to-end PMTU feedback themselves. Therefore the implementation must configure usable path MTUs and prove IP-layer feedback or appropriate edge sizing; it cannot assume that enabling MPLS makes PMTUD work. [S3] [S5] [S6]

The three-node qualification includes push/swap/pop, unknown-label drops, trusted-interface enforcement, TTL expiry, minimum/maximum packets, fragmentation policy, GSO on/off, TCP MSS behavior and required ICMP delivery. It captures labels on each hop and proves normal IPv4 delivery at the selected egress backend. Primary and repair paths are tested with their maximum overhead. Tests also verify that label changes do not alter TCP addresses, flags or payload.

Phase 3 adds transaction faults and two-generation coexistence. Phase 4 adds measured repair and admission. Phase 5 uses bypass-only mobility. Phase 6 separately validates COSTK's required interception and owner-steering hooks, including traffic reinjected after MPLS termination. Passing basic push/swap/pop is necessary but does not prove any of those higher-level behaviors.

7.4 Software ownership and escalation policy

We write the collector adapters, GVC controller and constrained agent, schema validation, OpenWrt packaging/hotplug integration, accounting integration and reproducible tests. We reuse standard Babel/eBGP implementations and Linux forwarding. The baseline does not require a new custom MPLS kernel module, custom TCP option, IStack dependency or per-flow TCP state in transit routers.

If standard userspace coordination and forwarding cannot satisfy a measured requirement, record the failing case, target configuration, required behavior and safer alternatives before escalating. A matching maintained firmware upgrade or narrowly scoped vendor/upstream backport may be appropriate. ASIC driver work is conditional on real hardware capability. No team should start a replacement MPLS stack merely because a proposed abstract operation was unavailable on the legacy kernel.

The release decision is evidence-based: Phase 0 establishes capabilities; Phase 1 validates routing and collection; Phase 2 validates control logic in simulation; Phase 3 proves MPLS and transaction semantics; Phase 4 proves protection/admission; Phases 5 and 6 establish mobility and TCP ownership; Phases 7 and 8 qualify physical-grid and regional operation. Unsupported features remain explicitly disabled rather than being reported as completed by association with a later phase.

8. Stable-IP mobility and access

8.1 Scope and architectural decision

The initial mobility service supports managed IPv4 clients moving between enrolled access TBoxes while keeping their client address, home gateway, and Internet-facing NAT state. MPLS transports the original IP packet; it does not authenticate a mobile, allocate an address, transfer a TCP socket, or discover a new attachment. Those responsibilities belong to the access and mobility services described here.

The baseline uses a stable home anchor for both ordinary Internet forwarding and COSTK service steering. Every client-origin packet reaches that anchor before a bypass or proxy-owner decision is applied. Every retained COSTK owner's client-return packet also reaches the anchor before delivery to the current access TBox. Standard IP-in-IP tunnels identify these service paths without UDP; MPLS transports their outer IPv4 packets. This deliberately adds path length and an IP header, but makes the anchor's committed mobile binding the single delivery decision. A handover does not require synchronized delivery-pointer updates at every TBox that still owns an older COSTK connection.

This service is a proposed implementation, not an already verified capability of the current TBox firmware. Phase 5 qualifies addressing, access, and bypass mobility. Phase 6 adds COSTK admission and ownership-aware forwarding only after its relay and kernel prerequisites pass. The application still experiences finite interruption during some radio transitions; the requirement is measured connection continuity within the qualified outage and timeout limits, not zero packet loss.

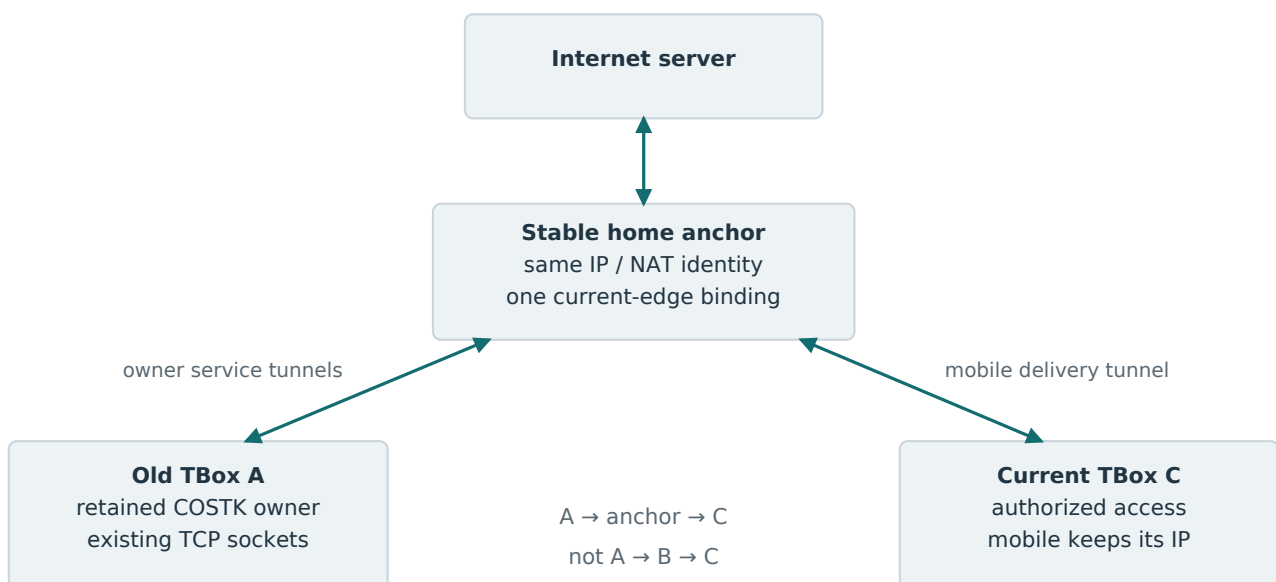


Figure 4. Existing COSTK flows remain at their original proxy. All owner returns resolve through the anchor, so successive moves do not create chains of old access TBoxes.

8.2 Address allocation and the home anchor

Infrastructure addressing and mobile addressing are separate. Allocation domains $T = 1$ through 253 use $10.T.N.0/24$ according to section 3.1: fixed node prefixes, routed links, loopbacks, outer service endpoints, inner proxy sources and local namespace links have distinct reserved N ranges. T is an IPAM identifier, not necessarily a literal town number. The mobile pool is $10.254.0.0/16$; infrastructure allocation must never use domain 254. The controller validates allocations for overlap before admitting a site.

An example mobility service reserves $10.254.16.0/20$ for a home anchor. Its isolated access segments present gateway $10.254.16.1$ with the same provisioned virtual gateway MAC. An example client receives $10.254.16.23$ with client netmask $/20$. A $/32$ in the routing design is a control-plane host route for that individual address, not an instruction to configure an ordinary phone with a $/32$ interface netmask. The address manager excludes gateway and other reserved addresses from allocation.

Item	Example or rule	Authority
Infrastructure	$10.T.N.0/24$, domains 1–253; N roles in section 3.1	Infrastructure address manager
Mobile service	$10.254.16.0/20$ within $10.254.0.0/16$	Home-anchor address manager
Client configuration	$10.254.16.23/20$, gateway $10.254.16.1$	Authoritative DHCP service
Delivery route	$10.254.16.23/32$ to committed serving edge	Mobility binding controller

An inter-grid move does not allocate a new address or substitute a nearer gateway. The target edge imports the existing service profile and forwards through the client's original home anchor. Address-space expansion and anchor migration require separate design changes; they are not implemented by reusing an infrastructure prefix or changing a live client's NAT gateway.

8.3 Stable identity and access authorization

Each enrolled client has a durable controller identity, an authorized address allocation, and a current attachment incarnation. A MAC address is an attachment attribute, not sufficient proof of identity. Production enrollment must bind a supported credential-backed identity to access authorization, for example through a qualified enterprise authentication or per-device credential mechanism. The exact AP/AAA integration is recorded in the capability manifest. A shared-PSK laboratory profile may demonstrate forwarding, but cannot claim to have passed individual-client impersonation resistance merely because its association reports are signed.

The access adapter joins authorized station state, enrolled identity, current MAC, tenant, and assigned address. Client MAC randomization or changed credentials trigger a verified identity-resolution procedure; an unfamiliar address is not silently associated with an existing subscriber. The tiny mesh reporter supplies observations, not authority to claim another client's address or move its binding.

Before activating data forwarding, the serving TBox installs source validation for the authorized client/address/attachment combination. DHCP bootstrap and required address-resolution traffic have narrow exceptions. Subscriber-supplied VLANs, labels, priority values, or service-context markers cannot grant access to another tenant or to COSTK return paths. Removal follows the attachment generation, so an old detach event cannot remove a newer client's rules.

8.4 DHCP authority, renewal, and rebinding

One logical authoritative DHCP service owns the anchor's client pool and persists leases independently of an access TBox reboot. All participating access segments relay to that service. Relay configuration maps a provisioned access-service identifier to the existing anchor pool; pool selection must not accidentally follow whichever physical infrastructure subnet the new relay occupies. Client-supplied relay metadata is discarded or replaced by trusted relay information.

The DHCP server identifier is a stable, reachable address of the authoritative DHCP service, not a changing access-TBox address. Unicast renewal reaches that server through the anchor path. Broadcast discovery, INIT-REBOOT requests, and rebinding are relayed from the currently authorized attachment. Server and relay behavior must be verified with the actual implementation, including reply delivery before the ordinary mobile binding is committed.

A valid lease remains valid when its client changes access TBox within the mobility service. The server must not reject it solely because the relay or physical access segment changed. A DHCPNAK is appropriate for a genuinely invalid allocation under authoritative policy, not as an automatic response to ordinary mobility. RFC 2131 describes renewal, rebinding, and address rejection; retaining a database entry without controlling those exchanges is insufficient. [S10]

The allocator prevents duplicate assignments and records lease expiry, identity, service, and reservation references. Expiry and address reuse are coordinated with connection and NAT cleanup so that a newly enrolled client cannot inherit stale delivery or proxy state. A client whose lease genuinely expires may stop using the address; continuity is not promised through unlimited DHCP-service failure. Phase 5 therefore tests renewal and rebinding during movement, server restart, temporary relay loss, and conflict handling—not only a download shorter than the lease lifetime.

8.5 Isolated access links, proxy ARP, and peer traffic

The common gateway IP/MAC exists on isolated access segments. The grid is not one stretched subscriber broadcast domain. Each segment needs client isolation, controlled DHCP/ARP handling, and a local gateway realization compatible with the vendor bridge and wireless driver.

Because clients receive a /20 netmask, they may ARP for another mobile in the same prefix. The access TBox supplies a policy-controlled proxy-ARP answer using the virtual gateway MAC for an authorized destination, then forwards the resulting peer traffic through the home anchor. The anchor applies tenant and peer-communication policy and resolves the destination's current binding. The same rule applies when two permitted clients happen to

be attached to one TBox; the initial implementation does not introduce a local shortcut that bypasses anchoring or isolation policy.

Proxy ARP must distinguish gateway resolution, remote-host resolution, and a client's own address-conflict probes. It must not manufacture a duplicate-address response against the legitimately moving client. The address authority and validated attachment determine whether a claim is legitimate. ARP tables, bridge forwarding entries, stale client-MAC learning, and duplicate-address detection are included in the access test fixture.

Host routes and neighbor entries are staged before activation. The new edge must be able to deliver to the actual authorized station or bridged leaf path, rather than simply contain a syntactically valid /32 route. No rule may expose a staged client address to subscriber data before the binding transaction permits it. M01, M02, M07, M08, and M10 cover the resulting attachment and isolation requirements.

8.6 Mesh-router qualification and deployment boundary

The memory-limited leaf retains its existing AP/EasyMesh firmware and the tiny C telemetry reporter. It does not receive COSTK, a GVC agent, MPLS modules, or a new heavyweight mobility daemon. Its existing firmware must nevertheless provide the access functions required by the selected profile.

For wireless bridged backhaul, both ends must support a matched, tested client-MAC-preserving transport, such as the applicable EasyMesh implementation or compatible four-address/WDS operation. An ordinary three-address station link is not evidence of transparent bridging for multiple downstream clients. Driver and firmware interoperability must be verified; enabling a setting named “bridge” is not an acceptance result. Linux Wireless documents the capabilities required at both ends of a four-address AP/station arrangement. [S12]

The bridge gate verifies at least three concurrent clients, independent source MACs and addresses, DHCP broadcasts and replies, ARP, tenant/VLAN treatment, and upstream reattachment without a loop. If that gate fails, per-mobile bridge mobility remains unsupported on that leaf. A routed-no-NAT profile may be considered separately only after stable host addressing, route reporting, neighbor behavior, and return delivery are proven.

A NAT CPE is a separate fixed-access profile. Its upstream identity does not reveal a trustworthy per-phone flow identity. Whole-CPE continuity additionally depends on preserving the CPE's own downstream addressing and NAT state. It must not be advertised as individual-phone mobility or per-phone adaptive COSTK merely because it reports station RSSI.

8.7 IPv6 and IStack disposition

IPv6 mobility is not part of the first release, and no IPv6 COSTK capability is claimed. If ordinary IPv6 access is enabled, it follows a separately specified bypass policy and must not circumvent tenant isolation. Its mobility acceptance remains unsupported until a profile defines router advertisements, default-router identity, prefix and address lifetimes, on-link flags, neighbor discovery, duplicate-address detection, and multiple active temporary addresses. IPv6 address allocation alone does not establish on-link behavior. [S11]

IStack remains an optional, separately evaluated comparison, never a dependency of the baseline. A failure to build, kernel crash, leak, packet corruption, or failed continuity/security/resource gate requires its rejection or disablement for deployment. This is a qualification policy, not a claim that every IStack operation is universally broken. The standard-kernel and userspace design must pass with IStack absent. No new kernel TCP-sequence translation module is required by the chosen stable-address architecture.

9. Handover control and recovery

9.1 Ownership and records

The mobility controller owns the desired client binding and handover transaction. The access-mobility agent owns verified local attachment integration. The GVC agent installs and verifies authorized circuit state; it does not choose an AP or authenticate a phone. The home anchor applies the committed delivery binding and retains NAT/firewall state. The existing Wi-Fi/EasyMesh control service remains responsible for supported radio operations.

The initial record separates controller leadership, client delivery generation, and AP event order:

```
MobileBinding {
    tenant_id, client_id, stable_ipv4, home_anchor_id
    serving_tbox, access_service, target_bssid, station_mac
    binding_generation, delivery_state, lease_reference
}

AttachmentEvidence {
    client_id, reporting_tbox, ap_id
    ap_boot_id, association_incarnation, event_sequence
    authorization_reference, observed_at, validity_deadline
}

HandoverTransaction {
    transaction_id, controller_leader_epoch
    expected_binding_generation, proposed_binding_generation
    previous_attachment, target_attachment, preparation_deadline
    access_result, circuit_result, anchor_result, operation_state
}
```

The binding generation is assigned by the authoritative control service, not independently invented by each AP. An authenticated report is still rejected if its boot identifier, association incarnation, or event sequence is obsolete. Controller fencing prevents two leaders from committing competing bindings. Packet forwarding never relies on wall-clock timestamps alone to decide which attachment is newest.

9.2 Candidate selection

Selection begins with AP/BSSID-to-serving-TBox inventory. Candidates must belong to the authorized tenant/access service, offer compatible client security, have adequate resources, and possess a viable path to the home anchor. Because established COSTK owners remain anchored, reachability of those owners through the anchor is also evaluated when predicting preservation of their sessions. An unreachable owner does not justify silently changing its flows to bypass; new or unaffected flows may still use a viable attachment.

Candidate quality combines client-observed beacon/scan signal, current AP station counters, candidate airtime and load, and measured grid-path behavior. Every value carries measurement direction, age, units, and availability. Current uplink RSSI is not interchangeable with candidate downlink signal. Candidate retry statistics for an unassociated client are generally unavailable. An advertised PHY rate is not available application throughput, and a retry ratio is not automatically a packet-loss rate.

The controller ranks eligible candidates by calibrated estimated end-to-end goodput subject to service delay/load constraints. It retains the present AP when estimates are close. A configurable improvement margin must persist for a configured hold interval before an advisory roam is requested; minimum dwell and retry backoff limit oscillation. Hard failure bypasses the performance hold, not authorization or readiness checks. Phase 5D initially runs this logic in observation-only mode and validates explanations against recorded traces before steering is enabled.

9.3 Wi-Fi preparation and client choice

The controller recommends a target, but an ordinary Wi-Fi client ultimately chooses its association. Supported 802.11k mechanisms help discover or measure neighbors; 802.11v BSS transition management communicates a recommendation; 802.11r Fast Transition reduces authentication work in a compatible mobility domain. None independently preserves the client's IP address or proves successful attachment. OpenWrt's roaming documentation is a capability reference, not proof that the captured vendor build supports every mechanism. [S13]

For a move between EasyMesh cells within the same 5×5 grid, request steering through the current cell's authorized AP-control adapter and verify target state through its own cell owner. Configure one steering authority path so autonomous EasyMesh recommendations and the mobility service cannot issue competing requests. Cross-cell control reachability, credentials and FT transport are explicit E08 gates; a common Babel domain or matching SSID does not satisfy them by itself. Cell membership is metadata, not a reason to change the client's home anchor or existing COSTK owner.

Preparation verifies SSID/security compatibility, mobility-domain configuration, and the supported FT key-holder relationship. FT-over-the-air is qualified with the actual clients first. FT-over-the-distribution-system is optional and requires a working, secured AP/key-holder control transport across the routed topology. MPLS IP reachability does not automatically transport every hostapd distribution-system exchange. Keys use supported protected mechanisms, not telemetry messages or improvised clear-text copying. [S14]

“Make before break” describes network preparation. It does not promise two concurrent working associations on a conventional single-radio client. The radio can pause while changing channels and reassociating. A client that lacks FT or ignores steering can still use ordinary client-driven reassociation within the stable-address service, with a separately measured interruption profile. Default steering does not forcibly deauthenticate clients to conceal missing support.

9.4 Preparation, attachment, and commit

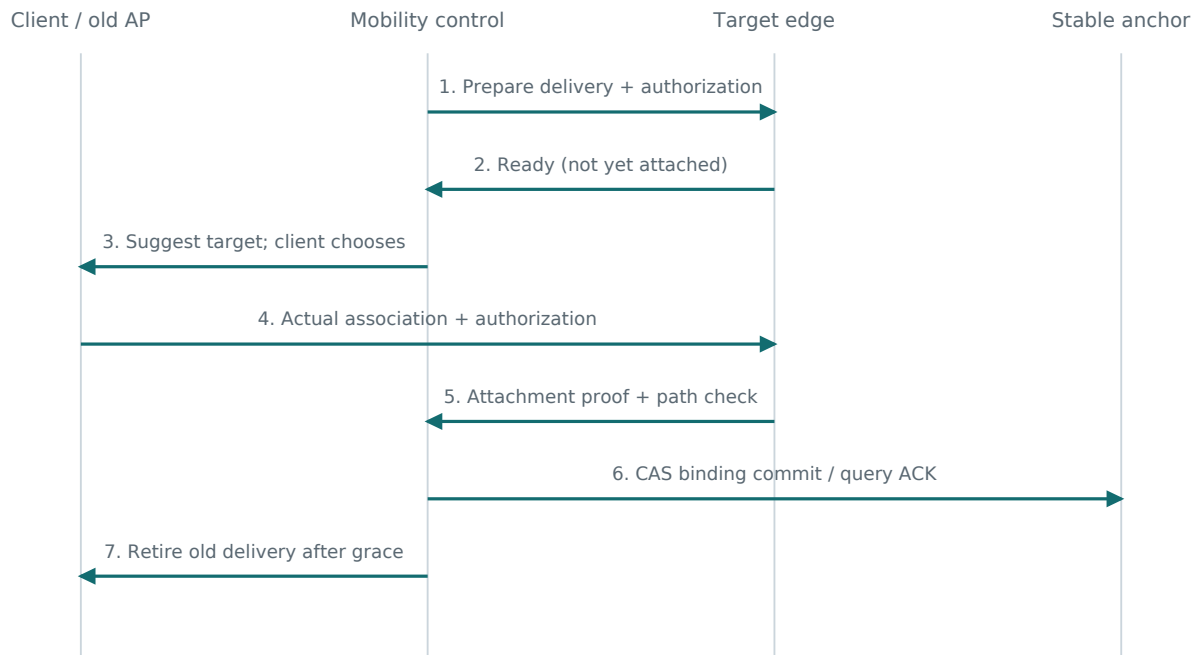


Figure 5. Predictive handover sequence. If the client attaches elsewhere, control follows that authenticated attachment. Preparation does not imply dual association on a single radio.

The predictive transaction stages the target access service, source validation, gateway/neighbor behavior, bidirectional circuits to the anchor, and trusted service-context handling before sending a roaming recommendation. GVC paths are unidirectional; a valid downlink does not establish a valid uplink.

READY means that staged network state has been read back and probed in a controlled test context. Before reassociation, the target cannot truthfully claim it already delivers to the mobile. ATTACHED means the adapter has checked current authorized station state and local protected delivery readiness. A BTM response, RSSI measurement, or application ping alone is not sufficient evidence.

At the home anchor, cutover performs a compare-and-swap against the expected binding generation. The proposed binding becomes authoritative only after target readiness and actual attachment are accepted. Subscriber forwarding at the new edge remains gated or bounded until the transaction permits it. Necessary authentication, DHCP, and validation exchanges retain narrow control exceptions so that preparation does not deadlock itself.

```

prepare(target, expected_generation)
verify_staged_access_and_both_circuit_directions()
if predictive:
    request_supported_roam_recommendation()

actual = verify_current_authorized_attachment()
if actual != prepared_target:
    prepare_and_verify(actual)

anchor.compare_and_commit(expected_generation, actual)
read_back_authoritative_binding()
activate_target_delivery()
retire_unused_preparation_and_bounded_old_delivery()
  
```

An atomic binding decision is not an atomic update of every switch in the network. Downstream forwarding is prepared first; the anchor then changes its one active delivery

mapping. A lost acknowledgment is resolved by reading committed state, not by assuming failure and reversing the transaction. During the short activation interval, bounded queuing or explicit loss may occur. Neither duplicate payload transmission nor zero-loss handover is part of the baseline.

9.5 Reactive movement and movement classes

If the mobile moves without preparation, the transaction starts from its verified actual attachment. If B was recommended but the client selects C, the controller prepares C and retires unused B state. An association at the old AP cannot be restored by merely restoring an old database value.

Event	Required change	State retained
AP changes under one TBox	Refresh local attachment and delivery	Anchor, address, owner records; usually circuit endpoint
Access TBox changes	Commit new anchor delivery binding	Address, NAT state, established COSTK owners
Client enters another grid	Prepare cross-grid paths to home anchor	Original home anchor and session identity
Mesh changes upstream TBox	Validate subtree, then move affected host bindings	Client associations where firmware preserves them

A pure grid-path failure is different: repair the path to the same access and service endpoints without generating a fictional Wi-Fi roam. A mesh subtree move can affect several clients at once; the controller tracks every affected binding and reports partial completion. It must not declare the whole subtree moved because one phone answers a probe. Regional-scale behavior is promoted only after the Phase 8 gate; Phase 5 uses a bounded inter-grid fixture.

9.6 Failure, fencing, and recovery

An unused preparation has a short lease. An old delivery path has a separate, bounded grace interval. A live COSTK owner's service path has a connection-lifetime reference and is not reclaimed by either handover timer. These lifetimes are independently configured and observable.

Failure	Required recovery	Forbidden shortcut
Preparation fails before roam	Abort candidate; retain verified old delivery	Expose partially prepared target
Mobile already left old AP	Complete or repair actual attachment	Treat database rollback as radio rollback
Commit reply lost or leader restarts	Fence conflicting writes; read and reconcile	Blindly replay or reverse a commit

Failure	Required recovery	Forbidden shortcut
New attachment cannot be reached safely	Report outage and bounded recovery	Remove isolation or proxy-owner policy

Agents reject stale leaders and obsolete binding generations while retaining verified forwarding. Controller-loss behavior does not promise arbitrary new mobility without an authority: installed delivery can continue, but a move requiring a new authoritative binding may incur outage until control recovers. Read-only probes and reconciliation must remain available without authorizing competing commits.

IP/Babel fallback is allowed only when the qualified fallback preserves the home anchor, tenant isolation, and required COSTK service context. A label miss cannot silently become ordinary Internet forwarding. If no preserving fallback exists, the service fails explicitly. Gateway failure that loses NAT state and proxy failure that loses sockets are stateful endpoint failures, not ordinary path failures.

9.7 Phase 5 evidence and release boundary

Phase 5A records capabilities; 5B establishes stable DHCP/gateway/ARP/source validation; 5C implements reactive transactions; 5D adds observable candidate ranking; 5E qualifies preparation and supported roaming; 5F adds recovery and subtree/inter-grid fixtures. COSTK admission remains disabled throughout this baseline.

M01–M12 cover local and cross-TBox movement, repeated handovers, capability fallback, ignored steering, stale metrics, pre/post-roam failure, replay/AP reboot, controller restart, subtree movement, anchored inter-grid delivery, and resource/MTU pressure. The full acceptance matrix defines stimuli and pass criteria. DHCP renewal/rebinding, address conflict, and multi-client source-MAC preservation are mandatory fixture checks, not optional demonstrations.

Evidence separates preparation, radio association, binding commit, and application outage; records loss/reordering/retransmissions; and reports p50/p95/p99 with resource bounds. A passing aggregate throughput number does not excuse an address change, unauthorized binding, loop, or application reconnect hidden by the test harness.

10. Adaptive COSTK and TCP continuity

10.1 Treatment is a new-connection decision

Adaptive COSTK chooses treatment only for a newly admitted eligible IPv4 TCP connection. Access telemetry can select COSTK using the qualified Westwood client-facing socket configuration or bypass using the original sender's congestion-control algorithm. CUBIC is the sender configuration in the controlled comparison, not an algorithm that a bypassing TBox can impose on an Internet server.

Established bypass remains bypass. Established COSTK retains the same proxy process and its two TCP sockets. A policy change, changed RSSI, new serving TBox, or new circuit generation does not change that ownership. COSTK publishes lifecycle facts and consumes policy/telemetry; authorized forwarding helpers install network state. It does not write Babel metrics, select a different home gateway, or optimize GVC routes.

No transit or backhaul COSTK proxy is enabled. An original access TBox can remain the owner of its existing sessions after the client moves; that does not authorize it to intercept unrelated transit traffic. Its retained client TCP leg now spans the anchor/grid path as well as the access link, so the former “Wi-Fi-only leg” description no longer applies. That additional path and its Westwood performance must be measured.

10.2 Anchor-centered service forwarding

The selected service backend is standard IPv4-in-IPv4 encapsulation, IP protocol 4, as specified by RFC 2003. [S15] Dedicated local/remote outer-endpoint pairs identify the following contexts; tunnel interfaces and receive-direction rules turn those identities into local forwarding policy. A local Linux packet mark is not transmitted across MPLS and cannot by itself carry identity between TBoxes.

Context	Sender and receiver	Anchor/edge action
CLIENT_IN	Serving edge → home anchor	Validate client; lookup committed bypass/owner decision
OWNER_INPUT	Anchor → selected COSTK owner	Deliver only to that owner's qualified interception path
SERVER_OUT	COSTK server leg → anchor	Apply anchored firewall/NAT; send toward server
SERVER_IN	Anchor → COSTK server leg	Deliver reply to existing owner socket; never TPROXY
CLIENT_RETURN	COSTK client leg → anchor	Resolve current mobile binding; never re-admit or re-proxy
CLIENT_DELIVERY	Anchor → current access edge	Deliver to authorized client; bypass interception

The access/anchor endpoint pair carries CLIENT_IN and its reverse CLIENT_DELIVERY. A separate anchor/owner pair carries OWNER_INPUT; an owner/anchor server-leg pair carries SERVER_OUT

and reverse `SERVER_IN`; a further dedicated pair carries `CLIENT_RETURN`. Endpoint addresses are allocated from reserved infrastructure service addresses, not client addresses. These are aggregate service tunnels, not one tunnel per TCP connection. The initial IPv4 profile prohibits overlapping subscriber addresses; tenant ACLs remain mandatory.

The selected isolation prototype places anchor client-service/admission and server-service/NAT processing in separate network namespaces, and runs each owner's COSTK service in a dedicated proxy namespace. It requires verified `CONFIG_NET_NS` support. Outer transport remains in the infrastructure routing context; constrained interfaces connect the service contexts. Both COSTK sockets stay in the owner's proxy namespace, with an explicitly different server-leg source tuple. Packet marks select policy but do not provide conntrack isolation. Phase 6 must prove namespace/hook placement, asymmetric replies, and identical-tuple collision rejection before this backend is supported; missing namespace capability requires a qualified replacement, not removal of isolation.

Before decapsulation, an outer-ingress ACL permits only the provisioned endpoint pair, protocol 4, and trusted infrastructure ingress. After decapsulation, verified tunnel-interface identity establishes the context and reapplies local policy marks; inner source validation follows. Subscriber ingress cannot inject these outer endpoint identities or supply a trusted context. IP-in-IP is neither encryption nor cryptographic authentication: the baseline requires a protected operator underlay, and crossing an untrusted network needs separately qualified protection.

At the anchor, decapsulated `CLIENT_IN` enters the admission/owner-lookup path before forwarding. At the owner, only `OWNER_INPUT` enters the transparent-socket/TPROXY rules and associated local-delivery policy route. `SERVER_IN` reaches its existing local server-leg socket normally. The owner explicitly marks locally generated client and server socket output into their respective tunnel policy tables. `CLIENT_RETURN` at the anchor bypasses admission and selects the current binding; `CLIENT_DELIVERY` at the edge bypasses interception and reaches the authorized station. Policy-mark allocation must coexist with existing COSTK marks rather than overwrite them indiscriminately.

Exact outer-endpoint routes take precedence over inner service-policy routing and resolve through the MPLS underlay, preventing recursive tunnel encapsulation. Reboot and packet-capture tests verify inner netfilter re-entry, tunnel-device identity, marks, local-socket delivery, and route readback on the vendor kernel. This packet-hook contract is a Phase 5/6 implementation gate, not an assumed property of a service label or VRF.

With one transport label, the service packet is link | MPLS | outer IPv4 | inner IPv4 | TCP. Its additional overhead is normally 4 bytes for the label plus 20 bytes for the option-free outer IPv4 header. Budget any further labels or link encapsulation separately. Configure service MTU accordingly and verify necessary ICMP/PMTU delivery; do not rely on fragmentation to conceal a black hole. Basic native MPLS IP forwarding remains available, but this mobility service intentionally uses IP-in-IP and still requires no UDP.

Server replies are associated with the anchored server-leg tuple and delivered to that same proxy socket. Bypass replies use the original mobile flow and current mobile binding. The forwarding implementation distinguishes tuple aliases, tenant, and context so that a proxy's synthesized client-leg packet cannot be mistaken for an ordinary new client connection.

For an established owner A and current edge B, the downlink client leg is owner A → home anchor → edge B → mobile; mobile-origin ACKs travel mobile → edge B → home anchor → owner A.

On a subsequent move to C, only the anchor's current-delivery lookup changes. A does not need a new direct pointer to C, and no A→B→C chain is constructed.

This choice consumes additional anchor bandwidth and can create detours even before roaming when owner and client are co-located. It is a correctness-first baseline with explicit capacity gates. Local shortcuts, direct owner-to-edge optimization, and anchor migration require a later consistency design and separate acceptance tests.

10.3 Flow identity and immutable ownership

The anchor's authoritative flow registry distinguishes a known bypass connection from a missing record. A missing record is not permission to choose bypass. Connection identity includes tenant, original/reply tuples, connection incarnation, and proxy process generation where applicable. Both COSTK legs and their NAT aliases are associated with one admission decision.

```
FlowOwner {
    flow_id, tenant_id, connection_incarnation
    original_client_tuple, client_reply_tuple
    server_leg_tuple, nat_aliases
    mode: BYPASS | COSTK
    owner_tbox, owner_process_generation
    admission_state, registry_generation
    serving_policy_version, serving_binding_generation
    client_leg_lifecycle, server_leg_lifecycle
    path_references, idle_policy, closed_at
}
```

Ownership metadata does not contain transferable live TCP sequence state. Tuple reuse must be handled through validated lifecycle/incarnation rules; a late packet cannot attach to a new connection merely because its five-tuple matches. Initial support covers ordinary client-initiated connections. Unsolicited inbound connections, simultaneous open, and TCP Fast Open require explicit admission-policy qualification before being enabled, rather than accidental handling as ordinary new outbound SYNs.

10.4 First-SYN admission before TPROXY

The proposed admission primitive is a bounded NFQUEUE path at the home anchor, before any packet can reach a selected owner's TPROXY interception. The captured legacy configuration has CONFIG_NET_IPIP, CONFIG_NETFILTER_NETLINK_QUEUE, CONFIG_NETFILTER_XT_TARGET_NFQUEUE, and CONFIG_NETFILTER_XT_TARGET_TPROXY unset. The matching vendor firmware must therefore be rebuilt with the required facilities and dependencies unless a new live inventory proves an appropriate image already supplies them. The captured configuration is not proof of the current running state. Standard-module enablement is not a new custom kernel stack.

The exact queue API, verdict/mark behavior, firewall integration, and performance must pass a capability prototype. If they cannot enforce the contract, Phase 6 is blocked or redesigned; neither NFQUEUE availability nor a modern nftables implementation is assumed.

Only initiating packets requiring a decision enter the bounded admission queue; established known traffic uses the installed fast path. The serving TBox publishes the authenticated access-policy proposal: client identity, current binding generation, policy version,

COSTK-or-bypass choice, eligible owner, measurement validity and expiry. The anchor validates that proposal against the current binding and resource/safety policy, recognizes retransmitted SYNs, commits a recoverable owner record, verifies required forwarding, and only then releases the held packet. It does not independently replace the serving TBox's RSSI policy. A stale or absent proposal requires bounded refresh or an explicitly permitted pre-commit bypass/rejection outcome. Queue length, bytes, per-client requests, global requests, and decision deadlines are bounded. Queue bypass/fail-open behavior must not let an undecided packet reach either the Internet or a proxy.

Before commitment, an unavailable COSTK owner or exhausted resource may cause an explicit bypass decision under policy, or rejection. After commitment, the same flow cannot silently change mode. Repeated SYNs reuse the committed or pending decision; they do not allocate extra owners. Timeout or queue failure drops the pending packet safely and records a reason. Subsequent TCP retransmission can retry the same transaction or receive an explicit pre-commit decision according to state.

The current COSTK manager observes connections at `accept()`, after the kernel has already handled the handshake. Adding a registry callback there is insufficient for first-SYN ownership. `accept()` lifecycle reporting must be joined to the earlier committed admission record. A bounded handshake-admission barrier is serialized with mobility cutover at the anchor; the current access binding is revalidated before release. C02 and C08 exercise retransmission, cutover, and recovery races.

The owner's initial SYN-ACK is also emitted before `accept()`. Configure the client-return routing context on the listening socket before it accepts traffic, using a qualified listener `SO_MARK/request-socket` inheritance path or an equivalently verified policy backend. A mark set only on an accepted socket is too late. Capture initial and retransmitted SYN-ACKs on the exact vendor kernel, including SYN-cookie/overload behavior if enabled, and prove that they traverse owner → anchor → current edge. Listener readiness is part of owner admission readiness; failure blocks new COSTK admission.

The new server leg has a second ordering requirement: bind its dedicated owner server-service source address/port, publish that tuple alias to the anchor, and obtain a ready acknowledgment before `connect()` can emit its SYN. Reusing the mobile's source address for this leg is excluded from the initial profile. A rapid server reply must not arrive before `SERVER_IN` steering exists. Failed setup closes pending sockets and records failure under the original decision; it does not convert an established client/proxy handshake into bypass.

Allocate that server-leg source from the dedicated `10.T.252.0/24` inner-source pool and assign its `/32` only inside the proxy namespace. Outer tunnel endpoints use `10.T.253.0/24`; they are not reused as inner socket addresses. Exact context routes pass `SERVER_IN` replies from the transport namespace to the owning proxy namespace, avoiding root-local address interception. The server-side firewall/NAT sees this registered owner tuple; the client still sees the original server tuple on its separate transparent client leg.

10.5 TCP flags and relay correctness

TCP SYN, SYN-ACK, data, pure ACK, SACK, window updates, probes, FIN, and RST follow the same committed ownership policy. MPLS transit does not acknowledge application bytes or terminate TCP. Each COSTK TCP leg has its own endpoint state; an ACK from one leg is not forwarded unchanged into the other leg. The endpoint validates resets, and the mobility layer does not fabricate or suppress them to disguise a failed handover. [S9]

There is an existing relay prerequisite. Code inspection of `src/proxy.cpp` shows `read EOF` calling `sibling->notifyClose()` and marking closure; the sibling input is then held. The splice EOF path uses the same pattern. This does not implement the required independent half-close behavior and is expected to fail the opposite-direction continuation case. This finding is not a claim that the new acceptance test has already been run.

Phase 6 must first qualify and repair relay half-close handling in both buffered and splice paths. When input EOF arrives, drain already accepted bytes, propagate the corresponding outgoing half-close with `shutdown(SHUT_WR)` when safe, and continue reading/forwarding the opposite direction. Final close waits for the relevant directions, buffered data, errors, and endpoint lifecycle. Registry retention alone cannot preserve a session whose relay has already closed the sockets. C03-C05 include no-roaming relay checks before repeating them during handover.

10.6 Lifetime, unknown traffic, and endpoint failure

Live ownership and its referenced paths remain until both relevant socket lifecycles and required closing-state protection permit cleanup. A first FIN, quiet connection, zero window, expired AP association, or completed handover grace period is not sufficient. Bypass records use qualified connection-tracking and inactivity policy; they cannot be silently deleted merely because the mobile changes edge.

NAT and firewall state at the home anchor must survive for the declared idle interval. Keepalive and timeout settings are measured against that policy. Forced state expiry or a maximum-lifetime cutoff explicitly ends the continuity guarantee. Address and tuple reuse follow cleanup quarantine appropriate to the validated endpoint/conntrack behavior.

Unknown established packets use a bounded authoritative lookup and optional bounded holding. On failure they receive an explicit drop/alarm/recovery outcome, never a guessed new proxy or automatic bypass. Registry restart reconciles committed decisions and live proxy process generations. A restarted process at the same IP is not the original TCP owner.

Loss of an owner process or owner TBox loses its unreplicated sockets; affected sessions are reported failed. Other bypass/new sessions can recover if their anchor and paths survive. Disabling adaptive selection stops new COSTK admissions and drains existing owners; killing those processes is a distinct destructive failure test. No live owner is evicted merely to meet a cache limit—admission stops before resource exhaustion instead.

10.7 Phase 6 implementation and acceptance

The implementation sequence is: qualify the relay and kernel interception prerequisites; define the authoritative registry and lifecycle IPC; prototype bounded pre-TPROXY admission; install context-preserving fast paths; serialize admission with anchor binding cutover; and validate cleanup, recovery, and resource bounds. The exact steps appear in the phase plan, including the half-close repair prerequisite.

C01–C12 exercise mixed-mode repeated movement, handshake races, ACK/window behavior, half-close, reset/tuple reuse, idle/NAT expiry, asymmetric delivery, unknown-flow recovery, owner failure, graceful disable, spoofing/resource exhaustion, and repeated three-client handovers. Each retained COSTK connection requires evidence of both tuples, owner process generation, socket lifecycle, NAT mapping, current attachment, and service-context counters.

Release requires passing Phase 5 first, then demonstrating Phase 6 without IStack. Captures must explain unexpected SYNs, RSTs, detours, or duplicate interception. Incorrect owner changes, unbounded pending state, false continuity claims, and cross-tenant forwarding block release even when aggregate throughput improves. Phase 7 repeats the matrix on the physical grid; Phase 8 qualifies regional paths and anchor capacity without silently adding stateful-anchor migration.

11. Software delivery, installation and operations

11.1 Delivery status and component ownership

The GVC and mobility components described here are **PROPOSED software**, not existing OpenWrt packages or demonstrated features of the COSTK repository. Existing COSTK and the tiny C link-telemetry reporter do not implement the proposed GVC transactions, stable-address mobility, first-SYN ownership registry or anchor-return steering. The implementation programme must produce these components and their evidence before enabling them for subscribers.

GVC means Grid Virtual Circuit. Its controller computes an aggregate forwarding contract; its agent translates the assigned part of that contract into validated local Linux forwarding operations. The agent is a userspace service, not the packet-forwarding engine. Standard Linux routing and MPLS forward packets after installation. The mobility controller decides which authorized access TBox currently serves a client; it asks GVC to prepare transport and asks the access agent to prepare local delivery. Neither the GVC agent nor COSTK independently chooses a new mobile attachment.

Component	Software work to deliver	Authority boundary
Topology and metric service	Multi-link inventory, lifecycle, measurements, route snapshots and capability provenance	Read radio facts; update only the explicitly authorized routing-metric interface
GVC controller and agent	Path computation, resource accounting, generation transactions, netlink programming and reconciliation	Controller assigns intent; agent writes only its authorized forwarding resources
Mobility controller and access agent	Stable leases, authenticated bindings, AP adapters, neighbor handling and attachment transactions	Change delivery only after authorization and downstream verification
Flow-ownership integration	Admission serialization, immutable owner records, lifecycle export and bidirectional steering	Preserve existing TCP endpoints; no migration of live socket state
EasyMesh cell integration	Firmware-specific ownership/scale manifest, supported controller adapter, bounded aggregation and planned migration tools	One cell owner per Agent/radio; no second regional radio writer or assumed built-in controller federation

The initial transport data plane is ordinary hop-by-hop MPLS with one active transport label at a time, or ordinary IP for the SOFT mode. Mobility service steering uses standard IPv4-in-IPv4 tunnels without UDP, with dedicated endpoint pairs/decapsulation interfaces for access-to-anchor and anchor-to-owner service legs. Native MPLS carriage of IP/TCP remains valid for transport traffic not needing that service context. A SOFT IP route follows the routing table directly; it does not imply that Babel maintains an MPLS LFIB. Segment

routing, arbitrary label stacks, service-label/VRF backends, ECMP fast repair and ASIC MPLS offload are deferred options requiring separate qualification.

11.2 Role-specific installation profiles

Install only the software required for the node's accepted role. A signed deployment manifest records hardware identity, firmware/configuration hashes, package versions, controller compatibility and the enabled service profile. Role changes require a new manifest and the corresponding acceptance tests.

Role	Required installation	Excluded by default
Memory-limited leaf mesh/AP	Existing supported AP/EasyMesh firmware and tiny C telemetry reporter	COSTK, MPLS, GVC agent and new heavyweight mobility daemon
Transit/Relay TBox	Radio management as applicable, Babel/eBGP by role, GVC agent and qualified MPLS support	Client interception and per-TCP-flow transit state
Client-facing TBox	Transit profile where needed, access-mobility agent and approved telemetry adapters; COSTK only after Phase 6	Independent gateway/NAT relocation and unqualified roaming commands
Controller/gateway	Appropriate control services, stable-address/owner stores and anchor forwarding/NAT	Unqualified automatic anchor migration or duplicate active writers

The gateway needs a constrained local forwarding agent whenever the design asks it to install GVC or mobility state. “Grid/Relay only” must therefore not be read as excluding an active gateway endpoint. Conversely, a pure remote controller that forwards no subscriber packets needs no MPLS data plane.

Treat fixed NAT CPE and managed bridged mobile access as different profiles. A NAT CPE may retain its household-address model, but this design does not promise individual phone mobility or phone-specific policy through that NAT. A managed bridge must preserve the source MAC and IP of each client across its upstream link. Three clients behind a bridge must appear as three validated identities at the serving TBox. An SSID or AP-mode setting alone is not proof of this property.

11.3 Kernel, package and API contracts

Use the exact target vendor kernel, toolchain, configuration and symbol versions. The captured legacy Edge TBox configuration has MPLS framing but lacks label routing and lightweight-tunnel infrastructure. Its full MPLS capability requires a matching firmware rebuild, not a forced foreign kernel package. Required standard options include CONFIG_MPLS, CONFIG_LWTUNNEL, CONFIG_MPLS_ROUTING, CONFIG_MPLS_IPTUNNEL and CONFIG_NET_MPLS_GSO, with dependencies resolved in that exact tree. Expected modular outputs include `mpls_router.ko`, `mpls_ip tunnel.ko` and `mpls_gso.ko`. No new MPLS stack or TCP-sequence-tracking kernel module is a baseline deliverable.

Enable MPLS input only on validated trusted interfaces after their creation. Cold-boot and interface-recreation tests must verify global label capacity, interface policy and ordinary IP fallback. Install a matching `ip-full` only when the installed utility lacks the required operations. A newer package name or matching `uname -r` string alone does not establish ABI compatibility.

The captured kernel also lacks `CONFIG_NET_IPIP`, `CONFIG_NETFILTER_NETLINK_QUEUE`, `CONFIG_NETFILTER_XT_TARGET_NFQUEUE` and `CONFIG_NETFILTER_XT_TARGET_TPROXY`. Phase 5 therefore requires matching-vendor IPIP support; Phase 6 requires the queue/TPROXY features and dependencies. Its namespace-isolated service backend additionally requires verified `CONFIG_NET_NS` support. Compile-time availability does not prove the intended packet-hook order: validate decapsulation, namespace boundaries, conntrack isolation, local marks, queueing and interception on the actual rebuilt firmware before enabling admission. A packet mark alone does not isolate conntrack state.

Version each controller/agent schema. Startup exchanges must state supported API versions, actions, maximum request sizes and resource limits. Reject incompatible writes before PREPARE; keep compatible installed state and report the mismatch. Rolling upgrades support an explicitly tested version window, not an assumption that unknown fields or commands can be ignored safely. Circuit, attachment and proxy-process generations have different meanings and remain separately named in every API.

11.4 Privilege, persistence and observations

Use unprivileged policy processes and a small constrained forwarding helper where the target supports this separation. A process with unrestricted `CAP_NET_ADMIN` in the forwarding namespace is not technically confined to one routing table merely because its API advertises a whitelist. Enforce the boundary through process design, restricted IPC, validated inputs and available platform isolation; test attempts to alter Babel, unrelated interfaces and client-access MPLS policy. COSTK must not obtain route-writing privileges as a side effect of admission integration.

Persist committed intents, generations, allocation ownership, client leases, binding epochs and recoverable flow-owner metadata. A snapshot does not preserve proxy sockets or NAT state unless separately qualified replication does so. After restart, query actual kernel and proxy state, reconcile against the authoritative record and reject ambiguous ownership. Never restore “live” TCP continuity from a stale database entry alone.

Agents retain the last verified forwarding state during a bounded controller outage according to service policy. Prepared but inactive generations may expire; live referenced owner paths may not be reclaimed merely because an ordinary preparation timer expired. Two controllers require a real quorum/fencing or lease authority. Two machines with independent increasing counters do not by themselves prevent split brain. Unknown leadership freezes new changes while retaining safe committed delivery where possible.

Every event records monotonic time, synchronized wall time with uncertainty, node/process incarnation, transaction ID and relevant generations. Metrics include installation latency, unknown-label drops, attachment age, admission queue occupancy, owner references, proxy socket counts and controller reconciliation outcomes. Standard target MPLS support may not expose per-label byte/drop counters. Declare each counter's backend and granularity;

use qualified interface counters, bounded rule counters and test captures where necessary. Never label aggregate interface traffic as an exact circuit counter.

11.5 Operational rollback and failure runbook

Before rollout, save the working profile and record which flows, anchors and owner processes must remain reachable. To disable adaptive COSTK, stop new COSTK admission first and allow existing owners to drain. Do not stop or upgrade an active proxy process under the name of a harmless policy rollback. To disable GVC, qualify replacement IP paths with the same isolation, stable anchor and owner waypoints before withdrawing ingress classification. Removing an unknown MPLS label in transit is not a safe generic fallback action.

Initial activation of the authoritative flow registry is a separate maintenance operation. The baseline quiesces new connections and drains the old bypass/proxy sessions before enabling registry-required classification, then starts fresh admitted connections. It does not adopt an unknown midstream packet as bypass. A future migration tool may import validated live anchor conntrack state as BYPASS, but that requires a separately tested bootstrap/reconciliation barrier and is not part of the initial rollout. Report any forced session termination explicitly rather than calling it transparent activation.

For uncertain COMMIT, freeze conflicting changes, query the gateway and every required edge, establish one authority and finish or reverse only a verifiably safe transaction. After a mobile has left A, rollback cannot assume A still delivers it. Repair the actual authenticated attachment or declare outage. After proxy loss, report its sessions as lost and permit new sessions only under a new process generation. After anchor/NAT loss, distinguish ordinary reachability recovery from preservation of old sessions. The final resort for a failed routed-radio coexistence gate is the validated EasyMesh-only configuration, with service interruption stated explicitly.

12. Phased implementation and acceptance

12.1 Programme structure and promotion rule

Stage A is engineering validation: namespaces and a small physical bench, covering Phases 0–6. Stage B is the physical single-grid pilot, Phase 7. Stage C is regional validation, Phase 8. Stage A requires three representative TBoxes for ingress push, transit swap and egress pop, a gateway/server endpoint, supported mesh/AP access and three concurrent clients. Endpoint roles may share laboratory hardware only when the resulting limitations are recorded. A two-box fixture cannot prove a three-node transit swap.

Dates are estimates to be developed after the capability inventory and work breakdown. Completion means accepted evidence, not elapsed weeks. Namespace success proves only the tested semantics; it does not qualify a radio, legacy kernel, firmware API or throughput target. Keep IStack absent from the baseline throughout. Any IStack comparison is a separate experiment with separate outcomes and no production dependency.

Stage	Phases	Promotion evidence
A: bench and emulation	0–6	Platform manifest; routing, MPLS, mobility and ownership tests; bounded resources
B: physical grid	7	Real-radio failure and mobility distributions, capacity and operations rehearsal
C: regional	8	Inter-grid anchoring, partition behavior, redundancy and declared gateway-failure outcomes

12.2 Phase 0: prove platform, radio and addressing contracts

Archive the working EasyMesh-only v1 configuration. Identify exact hardware/SDK/kernel combinations independently for the legacy bench and proposed MaxLinear product. Inventory radio/BSSID/interface ownership and verify that a routed adjacency can coexist with required EasyMesh onboarding, channel and link management. Verify MPLS EtherType carriage, peer identity and bidirectional reachability without creating an unintended bridge loop. Record whether multiple usable neighbor adjacencies can remain established simultaneously; an IP protocol cannot route across a radio link that is unavailable below it.

Build a MaxLinear field matrix for RSSI direction, per-link/per-MLO granularity, retry and failure denominators, counter wrap/reset behavior, rate meaning, airtime, queue measurements and probe support. Mark each field supported, unavailable or unverified with firmware/API evidence. Do not infer available capacity from PHY rate or substitute zero for an unknown measurement.

Finalize IPAM before enabling mobile service. Reserve 10.254.0.0/16 exclusively for the proposed IPv4 stable-mobile pool; prohibit town ID 254 in the geography-based subscriber scheme. Allocate per-town backhaul /31 links from 10.T.254.0/24 and loopbacks from 10.T.255.0/24, checking their capacity and every existing deployment for overlap. The pool is an operator proposal, not permission to overwrite existing addresses. Resolve any collision

and record the accepted allocation before testing. IPv6 mobility is deferred to its own addressing, ND and acceptance work.

Deliver the capability manifest, interface inventory, IPAM validation, package/build recipe and three-node MPLS capture. Gate outcomes are distinct: failed radio/L3 coexistence retains EasyMesh-only v1; proven Babel routing with failed MPLS permits qualified IP-only v2; functional MPLS missing performance/resource budgets remains laboratory-restricted until those budgets pass. Optional ASIC support does not rescue an unproven software or radio contract by assertion.

12.3 Phase 1: routed underlay and multi-link lifecycle

Implement stable node/link identities and Discover → Validate → Measure → Stabilize → Advertise. Associate telemetry with peer, direction, interface incarnation, timestamps and capability provenance. Validate peer and interface ownership before a discovered record becomes routable. Handle link disappearance, counter resets and stale observations without retaining false healthy neighbors.

Start Babel with fixed, documented costs on isolated routed adjacencies. Originate exact eligible prefixes, validate dual-exit default policy and constrain eBGP imports/exports. The radio metric adapter is introduced only after its field semantics and update behavior pass replay tests. COSTK decisions, circuit reservations and controller preferences never feed back into Babel metrics.

Reproduce the 5×5 graph in emulation and remove every edge and node. Include node 2 with alternate path 1 → 6 → 7 → 8 → 3; identify which links were already established and separately measure any radio reassociation. Test partition-specific prefix withdrawal, relay/default failure, malformed telemetry and simultaneous multi-peer changes. Deliver schemas, daemon configuration, lifecycle unit tests and traceable convergence timelines. Promotion requires independent routing with GVC and COSTK stopped; no guaranteed convergence time or hardware speed is implied.

12.4 Phase 2: GVC intent, path computation and transaction simulation

Implement topology collection, aggregate FEC identity, constraints, deterministic tie-breaking, local-label allocation plans and the controller state machine. Distinguish unprotected and protected results. A greedy primary-then-backup algorithm may miss a feasible disjoint pair; test against independently enumerated small graphs and report algorithmic limitations rather than treating every unsuccessful search as proof of impossibility.

Create an agent simulator supporting PREPARE, VERIFY, COMMIT, ABORT, RETIRE and state queries. Separate controller generation from wire labels. Use distinct downstream label resources for overlapping generations; a management epoch is not an extra field in an MPLS packet. Model leases, uncertain replies, leader fencing and safe label reuse. Install egress/transit resources before activating ingress.

Namespace traffic must demonstrate actual ordered forwarding, not only a computed node list. SOFT uses ordinary IP. Explicit circuits use the agreed single-label hop-by-hop actions. Reject unsupported encapsulation and table-capacity demands. Deliver model/property

tests, fault injection, API compatibility tests and reproducible traces; no target-TBox performance claim follows from this phase.

12.5 Phase 3: real MPLS agent and complete lifecycle

Implement the constrained netlink backend, target capability checks, installed-state readback and device/interface recreation handling. Separate desired from observed state. Program every path segment, verify a controlled probe, activate ingress last and retain the previous generation until safe retirement. Reject input labels on untrusted interfaces and unknown labels in transit. Service-aware fallback takes place at a controlled ingress with a verified IP path, not by assuming the kernel will remove an unknown label and route its payload.

Run three-node push/swap/pop across two cold boots. Inject rejection or timeout at each participant, lose COMMIT replies, restart agents and change controller leadership. Verify that packets do not enter incomplete generations and that restart does not reuse a live label. Test MTU, TTL and ICMP error behavior, checksum/GSO interactions, unsupported operations and counter limitations on the actual firmware. Deliver the package/profile, operational commands and evidence bundle. Promotion requires correctness and the declared bench resource budget, independently of offload.

12.6 Phase 4: protection, QoS and admission

Begin with measured userspace local repair: a qualified local failure signal triggers replacement of the forwarding action with a preinstalled, verified alternate. State explicitly that this is not kernel ECMP fast reroute or guaranteed ASIC-speed switching. Measure detection, notification, agent scheduling, netlink completion and recovered packet delivery separately. If a alternate radio must associate first, report that additional interval.

Account for shared radios, contention domains and half-duplex airtime, not only independent edge bit rates. Reserve primary and restoration capacity according to declared protection policy; include simultaneous failures permitted by the service model. Stage queue/policer changes with path generations and roll back resource holds after failed transactions. Use conservative soft admission until field capacity measurements justify commitments.

Test shared-risk failures, exhausted backup capacity, stale backup health, priority preemption and oscillating load. A circuit without a realizable diverse alternate is UNPROTECTED. Deliver repair distributions, admission audits and queue observations. Strict throughput guarantees and FRER remain separately qualified options; no implicit feature promotion follows from passing ordinary MPLS forwarding.

12.7 Phase 5: stable IPv4 bypass mobility

Build the AP/BSSID-to-serving-TBox inventory first. Implement stable IPv4 leases from the reserved pool, a consistent logical gateway, isolated access segments, proxy ARP/neighbor handling and authoritative anchor host routes. Configure dedicated IPIP endpoint pairs and decapsulation-interface ACLs; reject unexpected outer sources and wrong-interface arrivals. Verify DHCP renew/rebind and duplicate-address prevention across movement. A common gateway address alone does not preserve a client's lease or authorize its new attachment.

Implement authenticated attachment events, binding epochs and PREPARE/VERIFY/COMMIT/RETIRE transactions. Prepare the new delivery path without prematurely changing the anchor binding. Verify actual authorized attachment before commit; a steering response is insufficient. Add candidate measurement provenance, confidence, hysteresis and failure override. Introduce selection in observation-only mode, then supported 802.11k/v/r operations. A client may ignore the recommendation or choose another AP; reconcile the observed attachment instead of claiming control of the client.

Run M01–M12 with COSTK admission disabled. Include three distinct client identities behind the mesh and a mesh-upstream switch. Demonstrate inter-grid anchoring in emulation while keeping the original gateway; physical regional evidence remains Phase 8. Deliver the access adapter, binding database, addressing fixture, recovery runbook and long-lived bypass-TCP captures. Unsupported FT or NAT-CPE mobility remains explicitly excluded rather than recorded as a failed implementation of an enabled profile.

Package	Implementation order and deliverable	Evidence before advancing
5A: inventory	Map AP/BSSID, serving TBox, authenticated identity and firmware capabilities	Supported/unsupported/unverified features distinguished; three-client bridge identity verified
5B: stable access	Implement stable leases, gateway, proxy ARP, source validation and anchor service tunnels	DHCP renew/rebind and address-conflict behavior pass across prepared edges
5C: reactive transaction	Prepare actual attachment, verify both directions, commit anchor binding, retire old delivery	Unannounced movement and uncertain replies preserve one authority and safe recovery
5D: observed ranking	Add provenance, confidence, ranking, hysteresis and failure override without steering	Recorded traces explain choices; missing/stale data and oscillation handled
5E: predictive preparation	Stage network paths and use supported k/v/r operations before requested roam	Actual authorized attachment controls commit; ignored/rejected requests and client choice of C pass
5F: recovery	Add restarts, bounded cleanup, mesh-subtree movement and anchored inter-grid fixture	Complete M01–M12 and resource evidence; regional scale remains Phase 8

12.8 Phase 6: COSTK ownership and admission

First qualify and repair COSTK relay half-close in the buffered and splice paths, then implement lifecycle export distinguishing both TCP legs, tuple reuse and proxy-process restart. Prove correct no-roaming relay behavior before adding mobility. Define immutable flow mode and owner identity, including reply/NAT aliases. Register bypass flows too: “known bypass” must never be represented by “record missing.” Preserve original gateway/NAT and original proxy sockets throughout movement.

Prototype the first-SYN barrier at the stable anchor, before owner-side TPROXY interception, using target-qualified NFQUEUE support. Atomically establish the canonical

owner record at that anchor; bound queue size and wait time and test queue/process failure. Before releasing a SYN toward a proxy, commit recoverable ownership and verify the required steering. Serialize this decision with binding cutover and reuse it for retransmitted SYNs. If this cannot be made correct on the target, COSTK mobility admission does not pass; defaulting a midstream ACK to bypass is not a repair.

Keep adaptive access-policy ownership at the serving TBox. Its proposal includes client/binding identity, policy version, treatment, candidate owner, measurement validity and expiry; the anchor validates and atomically admits that proposal. Test stale proposals after a roam and permitted resource-rejection/pre-commit bypass behavior. The registry stores the admitted policy version for later explanation.

Before marking an owner ready, configure and prove client-return routing for the listening and embryonic socket states. Initial and retransmitted SYN-ACKs are emitted before accept-time socket configuration and must already follow CLIENT_RETURN through the anchor. Verify listener SO_MARK/request-socket behavior, or the qualified alternative, including SYN-cookie behavior when enabled. A correct post-accept data path does not pass this gate.

Add a second pre-connect gate for the proxy's new server leg. Bind its dedicated server-service source address and allocated port, publish that tuple alias under the committed flow incarnation, and wait for anchor acknowledgment that SERVER_IN return steering is ready before calling `connect()`. Do not reuse the mobile source address for this leg in the baseline. A near-zero-RTT SYN-ACK must find its preinstalled return context. Failed or timed-out setup closes pending sockets and records failure; it cannot retroactively turn an established client/proxy handshake into bypass.

Qualify the named namespace backend before these gates are enabled: separate anchor admission/client service from anchor server/NAT service, and use a dedicated proxy namespace at each owner. Verify CONFIG_NET_NS, interface placement and exact inner-packet hook order, including overlapping tuple collision rejection, conntrack isolation, queue marks, local socket delivery and locally generated output. Missing namespace support blocks this backend until a replacement isolation design is qualified; removing isolation is not a fallback.

Use the stable anchor for all owner-return delivery in the baseline. Access↔anchor and anchor↔retained-owner IPIP legs use dedicated endpoint pairs/decapsulation interfaces and trusted ACLs; a local firewall mark is not carried through MPLS. Assign local marks from validated tunnel context and reject subscriber-forged context. Test exact hook and conntrack-mark behavior so owner returns cannot be intercepted again. The anchor maps the stable client to its current edge, so repeated $A \rightarrow B \rightarrow C$ movement does not create old-edge chains. Serialize canonical flow admission and delivery-binding cutover at the anchor; the new edge receives its client-service binding and source-validation state, not a replica of every flow owner's records. Never install a label for each TCP connection.

Run C01–C12, then paired performance tests. COSTK uses Westwood on its configured client-facing socket. Bypass uses the actual remote sender's congestion-control algorithm; CUBIC is selected only for the controlled experiment. The retained proxy's client leg may span the grid after movement and must be measured as such. Promotion requires bounded state and correct ownership before any throughput benefit is considered.

Package	Implementation order and deliverable	Evidence before advancing
6A: relay and lifecycle	Repair/qualify half-close; export both-leg lifecycle, process generation and tuple incarnation	Buffered/splice no-roaming tests preserve pending bytes and opposite-direction data
6B: isolated forwarding	Qualify NET_NS, IPIP service contexts, hooks, conntrack and owner socket delivery	Wrong-interface/spoof/collision tests reject unsafe traffic without reproxy loops
6C: admission gates	Validate serving-edge proposal; first-SYN registry barrier; listener SYN-ACK context; bind/register server alias before connect	Initial/retransmitted handshakes, rapid reply, queue loss and alias timeout obey every ordering gate
6D: handover barrier	Serialize anchor admission with binding compare-and-commit; keep owner records centralized	A → B → C retains multiple original owners; new edge needs only its authorized client binding
6E: lifecycle failures	Implement recovery, safe drain, idle/NAT policy and bounded state retention	C01-C11 distinguish recoverable control loss from dead proxy/anchor state
6F: performance	Run C12 and paired three-client downloads only after correctness gates	Publish individual/aggregate results, fairness, uncertainty and resource limits without hiding failed flows

12.9 Phases 7 and 8: physical and regional qualification

Phase 7 rolls out one feature at a time on the physical 5 × 5 grid: qualified EasyMesh control profile, routed underlay, GVC, protection, bypass mobility and finally COSTK. Repeat radio/link/node/relay failures with simultaneous clients and mixed roaming capabilities. Qualify either the full 25-node single-controller envelope or accepted non-overlapping cells and their cross-cell adjacencies. The source's overlapping two-zone/row-3 arrangement remains a separate vendor experiment, not a required or presumed supported baseline. Rehearse rollback without destroying live proxy owners. Record available table, memory and CPU headroom under handover and recovery, not only steady load.

Phase 8 adds multiple grids, diverse relay/backbone paths and gateways. Test stitching, prefix isolation, mass recovery and anchored inter-grid mobility. The original gateway remains the session anchor. Without separately qualified stateful HA, gateway failure may lose existing sessions even when new sessions recover through another gateway. Test controller fencing and asymmetric partitions, not just clean process stops. Promote through limited canaries only after operators accept the failure semantics, numeric service budgets and evidence. A site without two genuinely diverse exits needs an explicit degraded-service exception.

12.10 EasyMesh scalability work packages

These packages implement ES1 for the 25-TBox grid. They augment the existing phases; they are not a new unbounded project or a claim that Babel itself implements EasyMesh scaling. Preserve the physical grid when evaluating a logical control split, and prove every required cross-cell connection rather than deleting inconvenient links from the test topology.

Phase	Required deliverable	Promotion evidence
0	Vendor limits and multidimensional scale manifest; complete 25-node radio/Agent ownership map; control and cross-cell transport prototype	E01-E03 inventory/prototype; identify unsupported single-domain or cross-cell arrangements before selecting a production profile.
1	Cell-aware link identities, controller scope and discovery isolation; qualified routed cross-cell adjacency	E03/E04: no accidental cell merger, conflicting radio owner or reliance on a region-wide control bridge.
2	Supported cell-controller adapter, versioned manifest admission, bounded telemetry queues and regional policy API	Replay E05/E10 event load; distinguish emulation from real radio/controller qualification.
5-6	Cross-cell access-control/identity/FT integration and existing anchor/COSTK rules	E08 together with M01-M12/C01-C12; no competing steering or implicit TCP-owner transfer.
7	Physical scale staircase and loaded recovery for all 25 grid TBoxes; planned split/migration runbook	E01-E09 at accepted peak, including a minimum 24-hour steady-load soak and separately scheduled recovery bursts.
8	Multiple accepted grids/cells, bounded aggregate management and failure containment	E10 plus repeated E07/E08; regional capacity and anchor behavior remain inside their separate limits.

The manifest includes `grid_id`, `cell_id`, `active_radio_controller_id`, approved Agent identities, radio/MLD ownership, controller discovery/interface scope, cross-cell backend, firmware hashes, accepted limits, measured headroom and policy version. This is proposed configuration schema. The adapter rejects inconsistent ownership or an unsupported vendor action before changing radio state. Low-memory leaf routers receive no new cell-orchestration daemon; capable TBoxes or management hosts provide these functions.

Scale testing advances through declared population steps to the intended load and the accepted boundary. Testing above a vendor-supported boundary requires a separately isolated laboratory authorization, not a production rollout. Do not infer production capacity from a fitted linear memory curve or an emulated Agent count alone. The report must identify the first limiting dimension, its safety margin and whether the 25-node physical grid is supported by one domain, several cells, or neither.

13. Verification specifications and release decisions

13.1 Evidence, timing and resource budgets

All tests below are specifications, not executed results. Store the topology, IPAM/profile, hardware/firmware manifest, source/build hashes, API versions, route/circuit/binding generations, application connection IDs and raw observations for each run. Preserve unsuccessful and excluded runs with reasons. Application reconnects invalidate a claimed preserved connection even if an iperf wrapper eventually reports throughput.

Use monotonic clocks for local durations and synchronized clocks with measured uncertainty for cross-host correlation. Capture enough packet and socket evidence to distinguish L2 reassociation, binding commit, MPLS repair, TCP retransmission and application pause. If capture overhead materially changes performance, repeat a matched instrumented correctness run and a lighter performance run; do not silently combine them.

Budget	Provisional engineering allocation	Required release treatment
Local repair	Investigate p99 below 50 ms only with an already usable alternate	Measure every component; this is not a guaranteed radio-failover SLA
Forwarding resources	Target at least 30% headroom at qualified peak	Count backup and overlapping generations, owner references and recovery bursts
Leaf reporter	Assign an explicit RSS/flash ceiling from the smallest leaf before deployment	Measure existing AP load plus reporter; no invented footprint result
Admission and control	Configure finite queue, lookup, candidate and transaction limits	Publish actual sizes/timeouts and overload outcomes before acceptance

These allocations are proposed planning values, not observed passes. Establish numerical outage, setup-latency and idle-retention limits with the product owner before testing. Correctness failures cannot be averaged away by a favorable percentile or aggregate throughput.

13.2 Phase 5 mobility acceptance matrix

ID	Stimulus	Observable pass condition
M01	Move between APs under one serving TBox; renew DHCP	Same client IPv4 and gateway; valid new association; no unnecessary MPLS endpoint change or duplicate lease
M02	Move A ↔ B during a download; renew/rebind lease at B	Same application/TCP connection and address; authorized attachment and verified delivery precede binding commit; both directions resume
M03	Perform 100 alternating handovers per supported profile	No incorrect owner, loop or address change; report all outage/loss/reorder samples; expired auxiliary state returns to baseline
M04	Exercise FT, non-FT and partial k/v/r support	Record actual method/capabilities and timing; unsupported functions stay disabled; no unsupported FT success claim
M05	Reject/ignore steering or let client choose C instead of B	Commit only the authenticated actual attachment; unused B preparation retires safely; no commit from BTM response alone
M06	Replay noisy, missing, stale and asymmetric measurements	UNKNOWN remains explicit; configured confidence/hold/dwell/backoff applies; hard failure bypasses delay, not trust checks
M07	Fail preparation or authentication before/after the radio move	Retain A only while reachable; otherwise repair actual authorized attachment or report outage, never fictional rollback
M08	Reorder/replay ATTACH/DETACH and reboot an AP	Old epoch/incarnation cannot remove or seize the current binding; spoofed identity is rejected and logged
M09	Restart controller at every state and lose COMMIT reply	One fenced authority; idempotent reconciliation of actual anchor/edge state; no abandoned live delivery or contradictory commit
M10	Put three clients behind mesh; move its upstream attachment	Three distinct source MAC/IP identities remain visible and leases stable; all bindings deliver correctly; no bridge loop. NAT-CPE profile is rejected for per-mobile claims
M11	Emulate inter-grid move while another gateway is available	Existing sessions retain original anchor/NAT; all return delivery reaches the actual edge; no silent anchor substitution
M12	Vary MTU; congest queues; exhaust preparations/buffers	Account for 20-byte outer IPv4 plus 4-byte MPLS label before other overhead; correct size/error behavior, no silent MTU black hole, ACK starvation or live-state eviction

Run long-lived mobility transfers for at least ten minutes, with scheduled movements, and repeat using three concurrent clients. A successful feature-negative test proves safe

rejection, not support for the rejected feature. Preserve packet evidence around every unexpected SYN, RST, DHCP change or binding transition.

13.3 Phase 6 COSTK acceptance matrix

ID	Stimulus	Observable pass condition
C01	Mixed bypass/COSTK flows through A → B → C → A	Original modes/process owners persist; owner returns traverse stable anchor to current edge; multiple owners coexist without old-edge chains or double proxy
C02	Retransmit handshakes during cutover; stop queue worker; use near-zero-RTT server	Owner committed before client SYN release; listener routes initial/retransmitted SYN-ACK before accept; server alias/return steering precedes connect; binding and queue races fail safely
C03	Send ACK-only, SACK/window updates and zero-window probes	Every packet reaches the proper TCP leg; packet captures disprove payload-only classification or control-packet bypass
C04	FIN half-close while opposite-direction data continues; buffered and splice modes	Pass without roaming before repeating during movement; drain queued bytes, preserve opposite direction and avoid premature socket/owner/path release
C05	Valid endpoint RST, TIME-WAIT/closing state and tuple reuse	Normal TCP reset handling; old owner/tombstone never captures a new incarnation; cleanup has authoritative evidence
C06	Idle/resume within policy, then force anchor NAT expiry	Qualified idle flow retains state; forced expiry is explicitly lost continuity, not a successful migration
C07	Asymmetric arrivals; wrong tunnel endpoints/interfaces; colliding tuples and forged marks	Qualify NET_NS/interface/hook placement and conntrack isolation; only authorized decapsulation establishes context; correct socket/waypoint and no reproxy/hairpin loop
C08	Unknown non-SYN packets after registry/cache restart	Bounded authoritative recovery or explicit timeout/drop; never silently create proxy or select bypass for an unknown established flow
C09	Restart proxy process or power off original owner TBox	Its lost sessions are reported failed; new process/IP reuse is not socket continuity; unaffected flows recover where reachable
C10	Disable COSTK selection with active sessions	New admissions use recorded bypass decisions; existing proxies remain alive until safe drain; process termination is a separate failure test
C11	Tenant spoofing, stale authority and full ownership tables	No cross-tenant delivery, unauthorized overwrite or live-owner eviction; admission fails before exposing an unowned proxy session
C12	Three simultaneous clients, mixed owners, 100 moves and repeated closures	Connection evidence explains each outcome; bounded RSS/slab/entries and references return after cleanup; paired throughput and fairness recorded

For each retained proxy flow, record both leg tuples, connection incarnation, process generation, socket lifecycle, anchor/NAT identity, current attachment and service-path

evidence. Lifecycle metadata is not a substitute for actual socket continuity. Test half-close with a purpose-built application; iperf throughput alone does not exercise every TCP control state.

13.4 Paired concurrent download performance specification

Use a deterministic seed to generate the agreed client-specific RTT/loss cases. Each case has one COSTK arm and one no-COSTK arm with identical client assignments and impairments. Randomize paired arm order in a recorded, balanced manner and repeat independent seeds or repeated trials to estimate variability. Keep all three clients downloading simultaneously; synchronize a future start time, log observed starts and flag a run whose overlap falls outside the declared tolerance. Use separate server sessions/ports as needed so one iperf server process does not serialize clients.

The selected measurement interval is **180 seconds after 10 seconds of warmup**, not an assertion that this run has occurred. Use fresh TCP connections for each arm, verify the intended mode/owner and server/socket CCA, and keep backhaul COSTK disabled in both. Download means server-to-client payload; an iperf client may require its reverse flag to request that direction. This is not an additional upload workload. TCP acknowledgments still traverse the return direction.

Name every impairment precisely. For a target total RTT, first measure baseline RTT, then add only the nonnegative difference, distributing configured delay over the two directions as recorded. Adding the full target delay in both directions would approximately double the intended increment. Define the requested loss percentages as independent **per-direction packet-loss probabilities** for this experiment, apply them once in each declared direction and label results accordingly; they are not an end-to-end combined loss target. Values such as 0.002% must not be passed as fraction 0.002, which means a different probability. Inspect counters and captures to detect duplicate application, NAT-based client misclassification and qdisc placement errors. Record whether ACKs and control packets receive the same impairment and ensure that segmentation/offload does not make the effective packet unit ambiguous.

For each common measurement window, compute aggregate throughput from the sum of received application bytes divided by that same duration. Also publish each client's result and Jain fairness, $(\sum x)^2 / (n \times \sum x^2)$, so a gain dominated by one client is visible. Report paired absolute and percentage differences, median and distribution, and confidence intervals obtained across independent paired trials/cases using a stated resampling method. Do not treat every one-second sample from one connection as independent replication. Separate fixed-placement performance from handover tests and disclose retransmissions, interruptions, server capacity, radio conditions and missing/invalid cases.

Measure anchor and shared-link traffic in addition to delivered throughput. With remote owner and edge, a COSTK download can cause approximately 2X payload transmit and X receive on the anchor's grid side for application rate X, before ACKs/headers/retransmissions. Compare those counters with the expanded service-path demand in section 6.2. This is a cost of the chosen anchor-centered backend, not an unexplained throughput loss or a reason to omit a traversal from admission.

13.5 EasyMesh 5 × 5 scalability acceptance matrix

E01-E10 are specifications for ES1, not executed results. Record controller and Agent firmware, physical versus logical membership, client/radio/BSS load, backhaul topology, message/byte rates, CPU/RSS/slab, queue peaks, disconnects and recovery distributions. Set numeric limits before a run; an unapproved limit or missing metric cannot produce a PASS. Three mobile clients remain useful correctness probes, but do not validate the full intended client-state capacity of 25 TBoxes.

ID	Stimulus	Observable pass condition
E01	Increase managed grid Agents through declared population steps; test full 25-node or approved cell profile	Supported, tested and admitted limits remain distinct; all 25 physical TBoxes are accounted for; over-limit enrollment is rejected or confined to authorized lab testing.
E02	Vary radios/MLDs, BSSIDs, VLANs and clients independently at accepted Agent count	Actual limiting dimension is identified; no hidden table exhaustion, stale client buildup or misleading node-count-only capacity claim.
E03	Exercise intended degree/depth and all required grid edges; identify local control-path articulation points	Inventory all 40 orthogonal candidate links where that topology is claimed; prove control and payload paths independently; no discovery leakage or forced foreign enrollment.
E04	Introduce competing controller discovery, partition control, restore connectivity and reboot Agents	One approved owner remains or controlled recovery is declared; no oscillating channel/SSID settings, accidental domain merger or stale-owner takeover.
E05	Burst Agent reboots, link changes, associations and metric requests across a loaded cell	Queues/resources stay bounded; resynchronization restores authoritative events; no guessed client location or coordinated endless rediscovery.
E06	Saturate payload traffic with telemetry enabled at accepted radio/client load	Controller, Agent and smallest leaf budgets pass together; no OOM/watchdog reset or control starvation; dropped metric samples become UNKNOWN.
E07	Fail a cell controller, uplink or middle TBox that cuts its control path while adjacent cells are loaded	Measure control recovery or qualified autonomous hold separately from a Babel payload detour; unaffected cells stay within budgets; no false control-recovery claim or regional restart cascade.
E08	Roam between cells and move a mesh upstream, with FT/non-FT clients	Actual attachment, stable address, original anchor and existing COSTK owner remain consistent; one steering authority; unsupported FT remains explicit.
E09	Execute approved cell split/Agent migration and inject failure at each step	No simultaneous conflicting ownership; required grid links and attachments recover; declared interruption/rollback is measured; live proxy owners are not silently killed.

ID	Stimulus	Observable pass condition
E10	Grow to several complete grids, then recover multiple cells concurrently or disconnect regional management	Local control stays bounded; summaries/replays respect aggregate budgets; no regional discovery flood or synchronized scan storm; mobility outage limits remain explicit.

For each profile, perform a minimum 24-hour loaded soak as an initial acceptance allocation, plus separately repeatable recovery bursts. This duration is a proposed floor, not a reliability guarantee. Report leaks, retries, stale objects and unexpected restarts, including any run that failed to finish. E07 and E10 distinguish loss of the regional policy service from loss of the GVC/mobility authority or a local EasyMesh controller; their failure effects are not interchangeable.

13.6 Final release decisions and unsupported claims

Approve only the tested profiles and failure envelope. MPLS forwarding success does not establish roaming, proxy continuity, airtime guarantees or hardware offload. A controller-selected target is not a guaranteed client association. A surviving alternate gateway is not preserved NAT/TCP state. Stable IPv4 is necessary for this baseline but cannot recreate a dead COSTK proxy. The absence of a custom kernel module is a design choice to reduce risk, not proof that all required userspace integration already exists.

Keep source-document ambitions distinguishable from engineering acceptance. Sub-5-microsecond switch action is not end-to-end radio failure recovery; a universal 50% throughput loss per wireless hop is not this design's capacity model. Do not assert that every 6 GHz operating mode uses DFS, or that a generic TSN switch removes Wi-Fi contention. Regulatory/channel obligations require the exact country, band, power mode and certified vendor configuration. COSTK's demonstrated split-TCP behavior must not be redescribed as arbitrary ACK suppression, RTT falsification or TCP-state migration without corresponding implementation evidence.

Release blockers include unsupported active interfaces, incompatible modules, route/label or bridge loops, stale-authority takeover, changed client identity in a retained session, double proxy, false success after owner/anchor loss, uncontrolled state growth and unqualified security boundaries. Record deferred performance features and unsupported access profiles separately from defects in enabled functions. The release packet contains signed decisions, raw evidence, accepted budgets, recovery commands and a named operational owner; it must remain possible to trace each claim back to a requirement, implementation artifact and executed test.

14. Decision record, open gates and references

14.1 Decisions that replace the earlier revision

This revision establishes one internally consistent starting point. The following decisions intentionally supersede ambiguous or incorrect statements in the previous presentation. They do not authorize live firmware changes; the corresponding phase must still qualify the selected backend.

Decision	Selected baseline	Reason and consequence
D01: GVC ownership	Proposed controller plus node-local userspace agent; standard kernel forwarding	The abstraction requires software integration, not a new MPLS protocol module.
D02: explicit paths	Hop-by-hop transport LSP; one label imposed initially	Avoids an invalid trailing Binding SID and unsupported long-stack imposition on the captured upstream baseline.
D03: SOFT forwarding	Ordinary service-aware IP with explicit service endpoints	Static MPLS LFIB entries do not automatically change with Babel's IP next hop.
D04: generation isolation	Distinct local incoming labels and authenticated management generations	The native LFIB does not key on an extra epoch or ingress-interface field.
D05: miss and repair	Unknown MPLS label drops; measured agent-driven repair or qualified ingress fallback	ECMP is not an assumed primary/backup mechanism, and arbitrary pop-on-miss is unsafe.
D06: service context	Standard IPIP endpoint pairs and isolated service processing	A service label or local packet mark does not automatically select a remote VRF or proxy leg.
D07: mobility	Initial IPv4 profile with authoritative DHCP, stable gateway and anchor binding	A stable /32 route alone does not define client subnet, ARP or lease behavior.
D08: retained proxies	All client and owner-return delivery resolves through the home anchor	Removes the need to update every old owner after every roam, at the cost of anchor traffic and detours.
D09: admission ordering	Anchor commits first-SYN ownership before interception; owner publishes server alias before connect	An accept-time callback is too late to protect the initial client handshake.
D10: proxy lifecycle	Half-close repair/prequalification precedes COSTK mobility	Existing EOF handling does not establish continued opposite-direction transfer.
D11: qualification	Standard-kernel rebuild and userspace work, then bench → grid → region	Correctness, resource and physical-radio evidence are independent release gates.

Decision	Selected baseline	Reason and consequence
D12: EasyMesh grid scale	Qualify the full 25-TBox controller envelope or use smaller non-overlapping control cells with proven cross-cell links	Preserve the physical 5 × 5 grid; Babel/MPLS does not remove EasyMesh controller, Agent or RF limits.

The initial service route may therefore carry more overhead and traverse more anchor capacity than a future optimized implementation. This is an explicit tradeoff: correctness is specified first, and each optimization must demonstrate that it preserves the same identity, authorization and ownership invariants. A later direct owner-to-edge shortcut needs its own multi-owner binding transaction; it cannot be added as an invisible routing optimization.

14.2 Commissioning questions and accountable owners

The following questions are bounded commissioning work, not unspecified architecture. A failed answer has a defined consequence. Record evidence and a named individual for each functional owner before commencing hardware rollout.

Open gate	Responsible function and required evidence	Failure disposition
Vendor radio/L3 coexistence	Platform/radio: actual per-peer interfaces, required EasyMesh control ownership and simultaneous data-path captures	Retain qualified EasyMesh-only v1; no unproven routed release.
EasyMesh 25-TBox scale	Radio/platform/test: complete Agent/radio/client/recovery envelope and E01-E10; supported non-overlapping cell interconnection where needed	No full-grid scale claim if neither single-domain nor cross-cell profile passes; obtain a vendor enhancement or separately approved management replacement.
Vendor metric fields	Radio/telemetry: exact API, units, granularity, freshness and reset behavior per firmware	Mark unavailable fields unknown; disable dependent admission/metric features.
Kernel forwarding backend	Platform/routing: reproducible matching build, MPLS/IP/IP/namespace and MTU tests	Upgrade/backport only by approved scope; restrict to a profile that passed.
Mobility identity integration	Access/security: DHCP authority, lease persistence, actual authorized association and FT key-holder transport	Bypass mobility unsupported on that access profile; do not claim pre-association success.
COSTK lifecycle and admission	COSTK/control: half-close, pre-SYN and pre-connect barriers, both socket tuples and restart behavior	Keep COSTK mobility disabled; qualified bypass mobility remains available.
Numeric service budgets	Product/test: peak clients/flows, setup and outage limits, control/leaf memory, throughput and retention policy	Laboratory status only; no unspecified guaranteed SLA.

Open gate	Responsible function and required evidence	Failure disposition
Anchor resilience	Gateway/operations: NAT and socket-state failure envelope, capacity under detours and retained owners	Declare affected sessions lost on anchor-state failure; do not promise seamless anchor migration.
Untrusted backbone protection	Security/platform: authenticated encryption profile and measured MTU/CPU/failover effects	Do not expose unauthenticated IPIP service contexts across an untrusted network.

14.3 Requirements-source claims not adopted without evidence

The supplied material is valuable for identifying the desired system, but some claims cannot be used as implementation assumptions. The design does not use a fixed fifty-percent throughput penalty per wireless hop: actual contention, radio reuse, interference, packet size and scheduling determine capacity. It does not equate a switch's local forwarding latency with end-to-end radio failure recovery. It does not use TTL as permission for a routing algorithm to create loops.

The design also does not assume that every Babel daemon supplies ECMP, that a label automatically activates hardware offload, that a TSN switch eliminates radio contention, or that two EasyMesh controllers can simultaneously own the same interface. Those are implementation and vendor qualification questions. Required regulatory behavior is determined by the certified country/band/power configuration; a universal statement about 6 GHz DFS is not used as the radio-control contract.

COSTK is treated as a split-TCP implementation with explicit socket and congestion-control behavior. Claims about arbitrary ACK holding, hidden loss, RTT manipulation or transparent migration of TCP state are not adopted as baseline features. IStack remains an optional unqualified comparison: if it fails the stated gates, it is rejected or disabled and the standards-based design proceeds without it. “Not qualified” does not mean “all possible implementations are impossible,” but it does mean “not a production dependency.”

14.4 Local evidence index

The editable chapter sources, assembled Markdown, printable HTML, PDF and generated build audit are distributed together. Source document hashes and build-input hashes are recorded with the output so a later review can identify the exact input revision. PDF validation confirms a document artifact, not a hardware result.

Evidence	Location or identity	Use and limitation
Requirements input A	Integration_Request_Final.docx	Eleven integration requests; R1-R11 in section 1.4.
Requirements input B	20260831_1. v1_v2 Grid_TBox_EasyMesh_Implementation_Plan.docx	Topology, v1/v2 proposals, addressing and vendor qualification questions.

Evidence	Location or identity	Use and limitation
Saved kernel configuration	build/edge-4.4.60/kernel.conf ig	Compile-time evidence for the captured image; not current live-device inventory.
COSTK relay EOF paths	src/proxy.cpp, read/splice EOF and notifyClose()	Code-inspection basis for the Phase 6 half-close gate.
COSTK acceptance path	src/manager.cpp, Manager::run()	Accepted socket becomes visible after kernel handshake; not a first-SYN barrier.
Existing interception setup	files/costk.init, TPROXY/conntrack-NEW rules	Existing classification is not the new authoritative pre-interception admission service.
Existing IPv4 adaptive scope	docs/design/ADAPTIVE_SOLUTION _DESIGN.md	Current adaptive scope does not establish IPv6 COSTK mobility.
Earlier presentation	archive/revision5/	Preserved historical revision, superseded by this engineering design.
Prior portrait design	archive/revision6/	Preserved before adding the explicit EasyMesh 25-TBox scalability design and E01-E10.

14.5 Primary technical references

Bracketed S-identifiers in the chapters refer to the sources below. Standards explain protocol behavior; they do not prove that a particular vendor build implements an option. The Linux source references are pinned to upstream 4.4.60 for reproducibility. An actual vendor tree can differ and must be inspected independently. Proposed algorithms, schemas, work packages and acceptance criteria in this document are our engineering choices, not text prescribed by these references.

ID	Primary source	Relevant scope
S1	RFC 8966 — The Babel Routing Protocol	Distance-vector operation, feasibility and routing behavior.
S2	RFC 3031 — Multiprotocol Label Switching Architecture	MPLS forwarding and unknown-label behavior, especially section 3.18.
S3	RFC 3032 — MPLS Label Stack Encoding	Label encoding, TTL and packet-size considerations.
S4	Linux 4.4.60 — MPLS internal.h	Upstream structures and MAX_NEW_LABELS limit.
S5	Linux 4.4.60 — af_mpls.c	LFIB lookup, forwarding, multipath and route dumps.
S6	Linux 4.4.60 — mpls_ip tunnel.c	Label imposition and lightweight-tunnel state validation.
S7	Linux 4.4.60 — MPLS Kconfig	Standard MPLS configuration dependencies.

ID	Primary source	Relevant scope
S8	RFC 9256 — Segment Routing Policy Architecture	Binding SID policy selection and expansion; deferred SR extension.
S9	RFC 9293 — Transmission Control Protocol	TCP control states and transport semantics.
S10	RFC 2131 — Dynamic Host Configuration Protocol	Lease renewal/rebinding and address-validity behavior.
S11	RFC 5942 — IPv6 Subnet Model	Address allocation versus on-link determination.
S12	Linux Wireless — iw and four-address operation	Capability and interoperability requirements for wireless bridging.
S13	OpenWrt — Wi-Fi roaming	Configuration and capability considerations for roaming.
S14	hostapd configuration reference	FT, mobility domain and key-holder configuration.
S15	RFC 2003 — IP Encapsulation within IP	Standard IPv4-in-IPv4 packet carriage; no UDP dependency.
S16	Linux kernel — MPLS sysctl documentation	Platform label capacity and per-interface input controls; verify legacy support.
S17	prpl Foundation — prplMesh	Implementation's Agent/Controller roles and platform abstraction; not evidence of a target TBox capacity.
S18	prplMesh — Configuration documentation	Controller tasks, metric/statistics configuration and client-state limits; vendor/version-specific values require separate qualification.

14.6 Terminology

Term	Meaning in this document
Access edge	Capable TBox serving a direct client or a supported leaf mesh AP.
Anchor / home anchor	Stable gateway service retaining client identity, NAT and the authoritative delivery/flow decisions.
Binding	Mutable mapping from a managed client identity to its current authorized serving edge.
Circuit / LSP	Directional transport contract and its installed label-switched path, normally shared by many flows.
EasyMesh cell	Bounded radio-management domain within a physical TBox grid; not necessarily a separate Babel domain or subscriber subnet.
FEC	Forwarding equivalence class selected at an authorized ingress.
GVC agent	Proposed node-local userspace installer/verifier for controller-authorized forwarding.
LFIB	Label forwarding table; native Linux lookup is scoped to a network namespace and top label.
Owner	Original COSTK proxy process holding the two TCP legs of an admitted flow.
PCE	Path computation function using a validated topology and constraints.
Service context	Trusted identity of a packet's processing direction, derived from qualified endpoint/interface handling.
Shared-risk group	Links or nodes subject to a common failure or capacity dependency.
SOFT	Ordinary service-aware IP forwarding, not a magically Babel-synchronized MPLS circuit.
TPROXY	Qualified transparent socket interception at the owning access-proxy service.
Unknown	A deliberately unavailable or unverified value, not a zero measurement or permission to bypass.